

《Automotive Configuration》

Lecturer : Dr. HU Zongjie / Associate Professor

Contact: zongjie-hu@tongji.edu.cn

Mid-term Exam

Venue: F102

Time: 3:30-5:05 pm

Date: April 13, 2023

第12章 车用新型动力系统

Chapter 12 New Powertrains for Automobiles

12.1 History of EVs

12.2 Battery Electric Vehicle (BEV)

纯电动动力系统

12.3 Hybrid Electric Vehicle (HEV)

油电混合动力系统

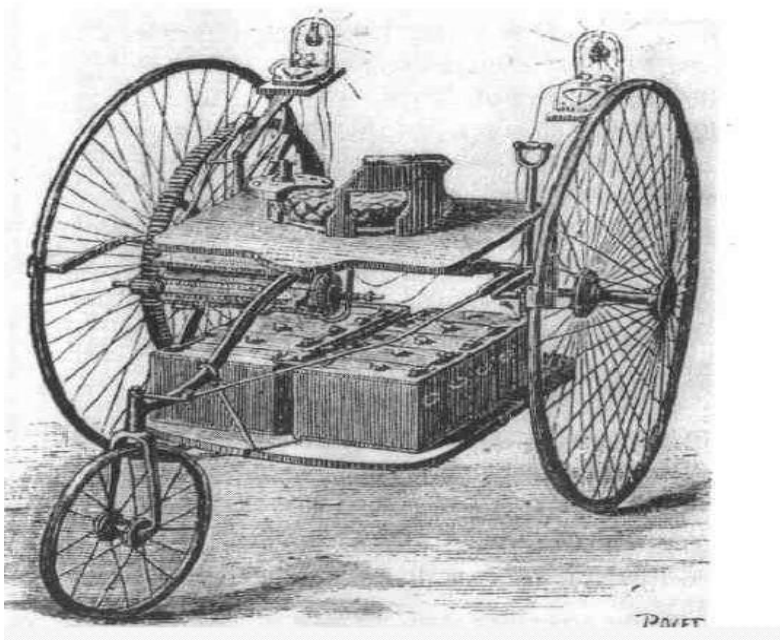
12.4 Fuel Cell Vehicle (FCV)

燃料电池（电电混合）动力系统

12.5 双碳背景下的新型交通“燃料”

绿色“燃料”

12.1 History of Electric Vehicles



The **first BEV** was invented by Frenchman Thomas Davenport in 1881—Lead Acid Batteries tricycle (铅酸电池三轮车). Then in 1882, Two English Professors Ayrton and Perry also assembled an EV; The world came to horseless Age, BEVs were being researched.



In 1886, Karl Friedrich Benz won the world's first patent of Gasoline Vehicle, marking **the first car** born in the world. From then on, ICE vehicles were competing with EVs.



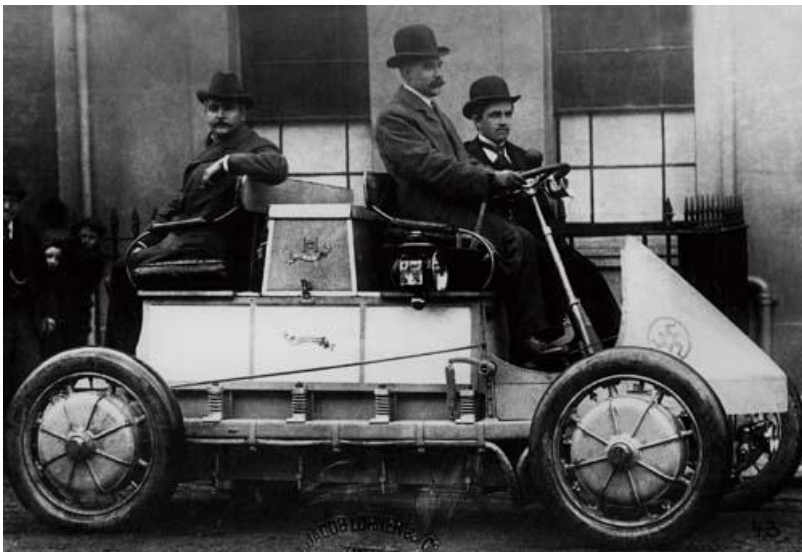
From the end of the 19th century to early 20th century, different power sources: Steam engine(蒸汽机), ICE and EV were competing seriously. But with the petroleum being exploited widely, and ICE's advantages, Steam engine and EVs were neglected by people.

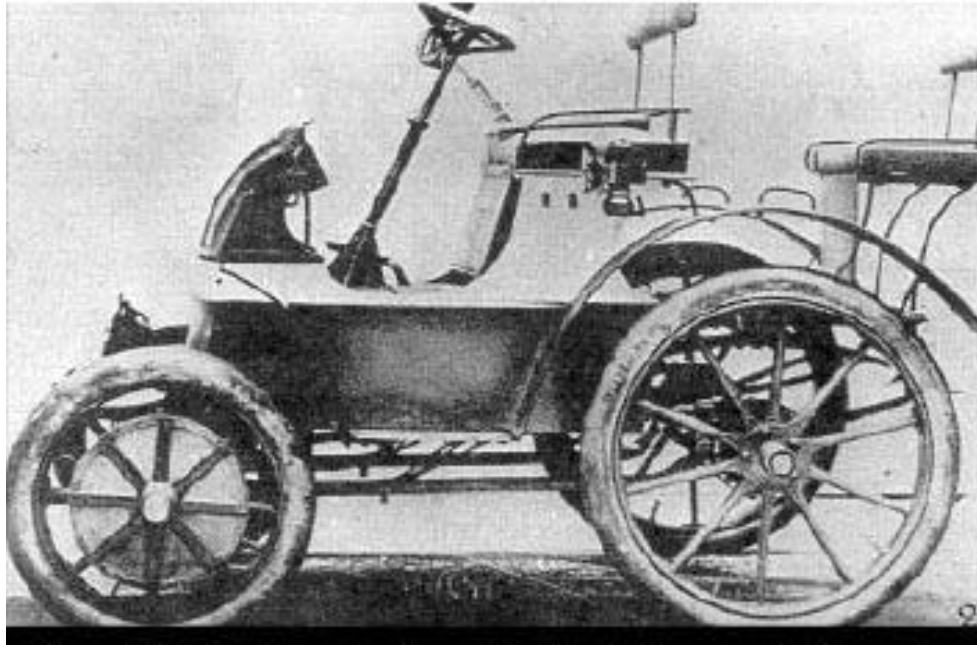
Why?

Long charging time.

Short driving distance.

1895~1905: Golden age of early EVs





- The Lohner-Porsche car (1897)
- Car designed by Ferdinand Porsche and built by Jacob Lohner
- Two hub motors
- A single gasoline engine powering a generator

The **Re-Electrification** of the Automobile: Trends and Motives in Hybrid and Electric Car Designs, Philip T. Krein

By combining 1008 cells, Harry Karl Ihrig, a farm equipment manufacturer in the U.S., produced a fuel cell stack, which could generate 15 kW and was capable of powering a 20 hp tractor, demonstrating the first fuel cell-powered vehicle.

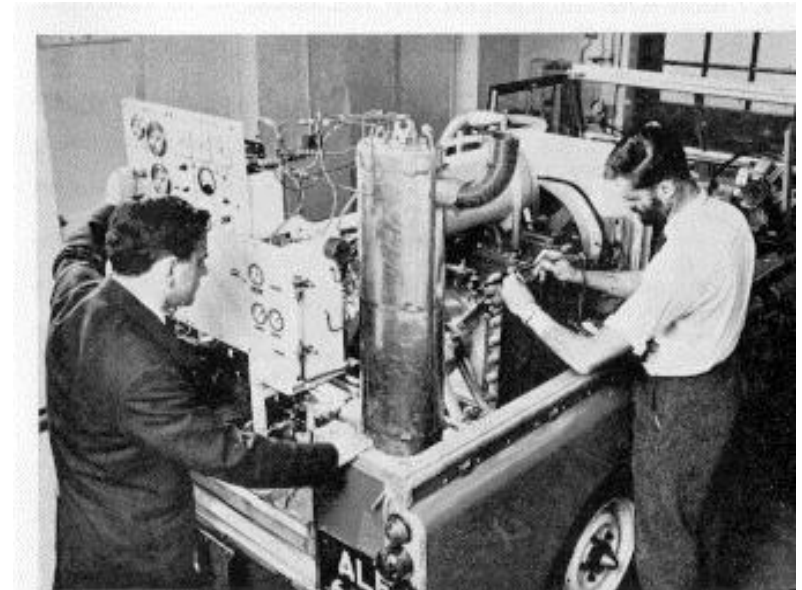


Fig. 111. "Shell" Research 5 kW hydrogen producer and fuel battery.

1966 Methanol Fuel Cell Truck

Electric Vehicle: use all or part of the electric energy to drive automotive

● **Advantages:**

1. Zero **local** emission
2. Low noise
3. High energy efficiency in application phase
4. More recyclable energy (braking/down the slope)
5. Improving energy structure
6. Less heat emissions

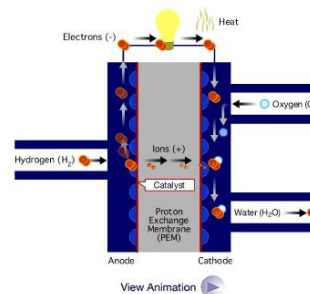
● **Disadvantages:**

1. Short driving range
2. High cost
3. New infrastructure
4. Source of Electricity/H₂

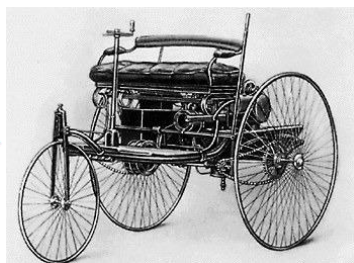
??电动汽车

??多元化

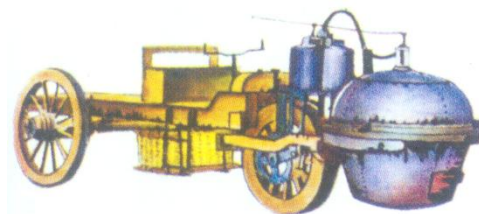
能源多元化



内燃机汽车



蒸汽机汽车



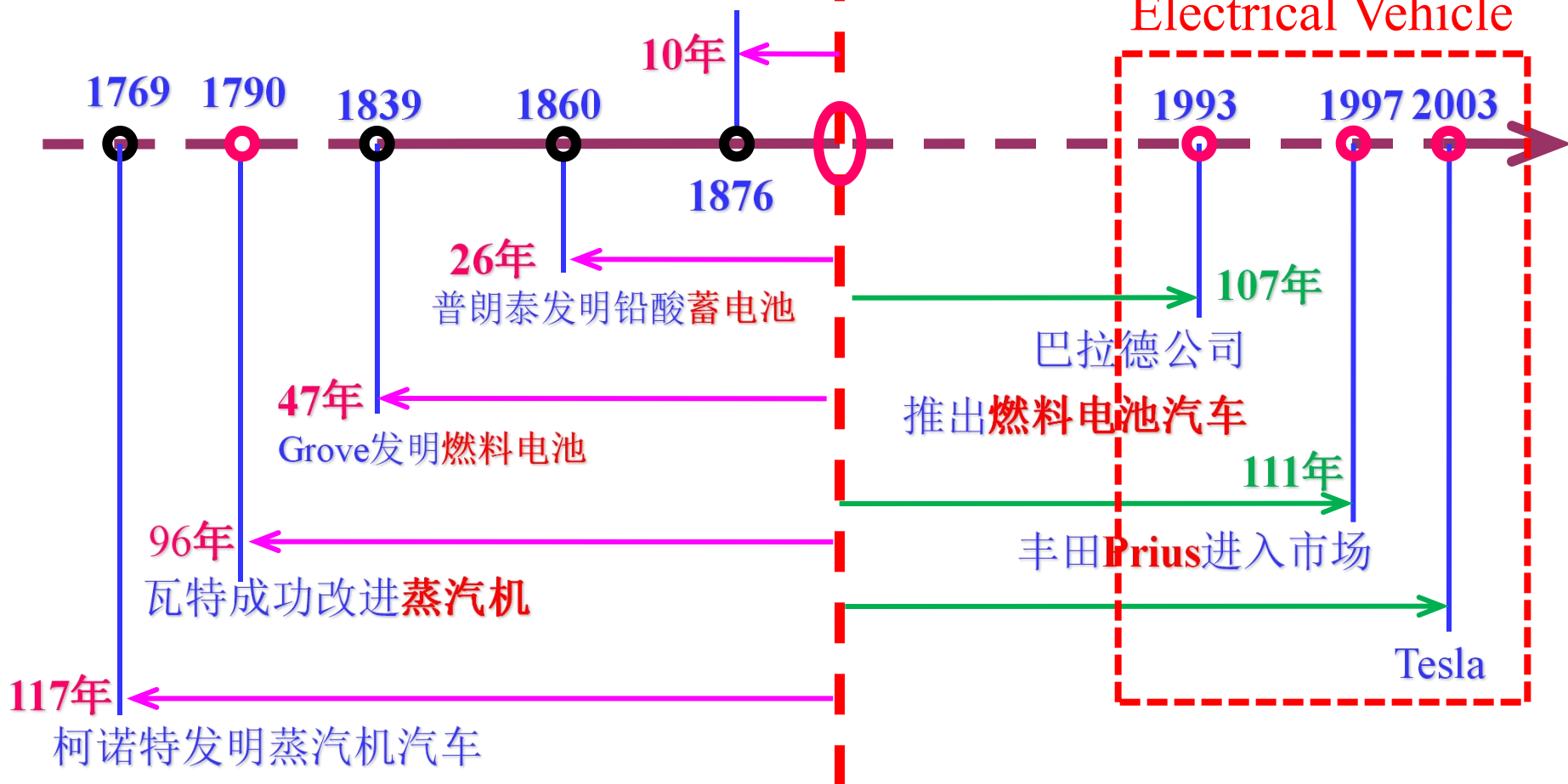


现代汽车元年

1886

NOT new energy vehicle

Electrical Vehicle



12.2 Battery Electric Vehicles

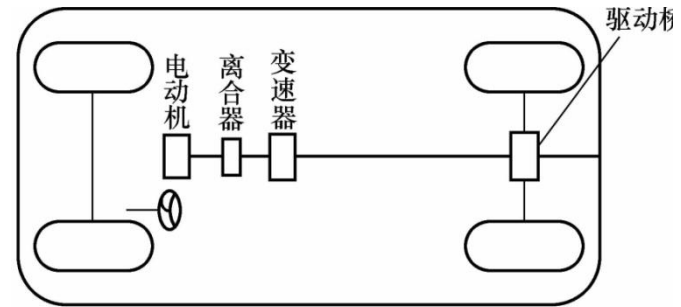


Elect

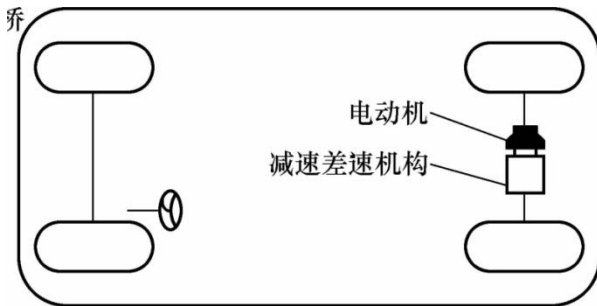
Types of BEV:

(1) Divided by components and arrangement:

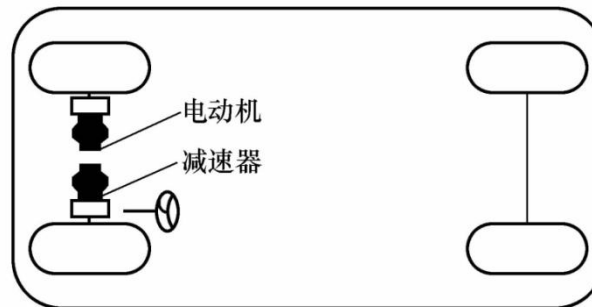
1. Conventional type
2. No transmission type(无变速器型)
3. No differential gear type(无差速器型)
4. Wheel hub motor type(轮边/轮毂电机型)



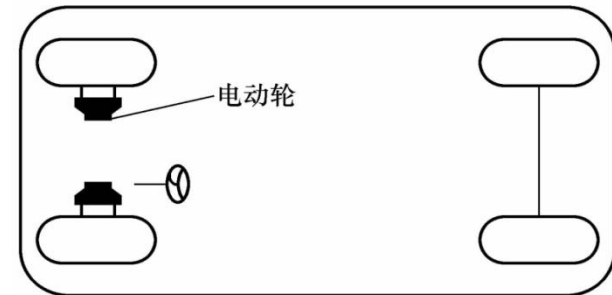
1. Conventional type



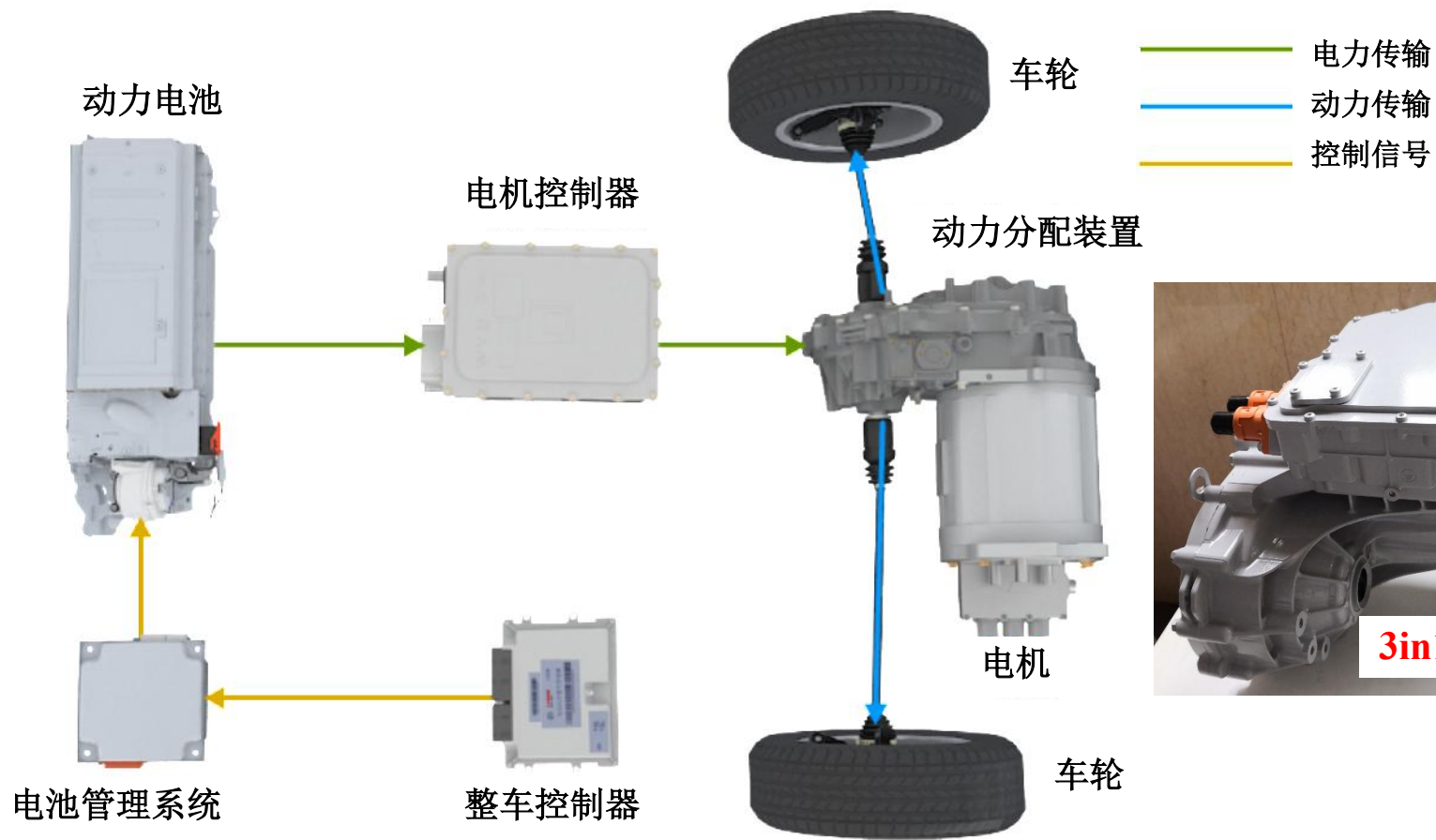
2. No transmission type



3. No differential gear type



4. Wheel hub motor type





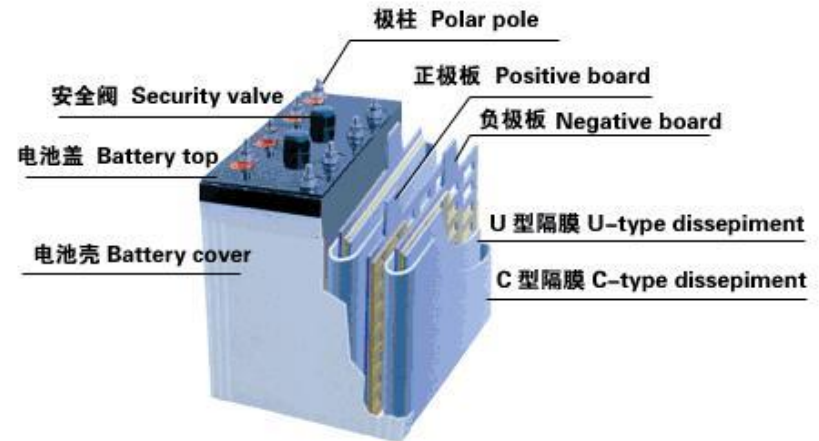
Wheel hub motor type

(2) Divided by types of batteries

1. Lead-acid batteries(铅酸电池)
2. Nickel-hydrogen battery(镍氢电池)
3. Lithium ion battery(锂离子电池)

(3) Divided by types of motors

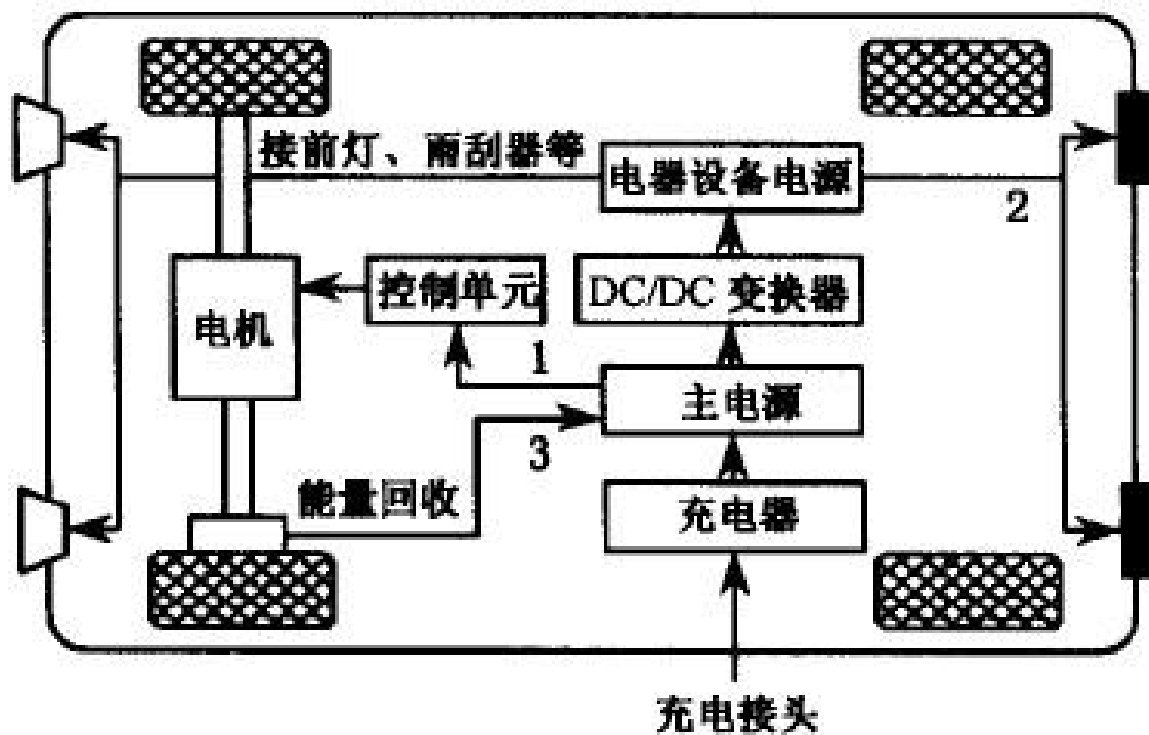
1. DC motor driving type
2. AC motor driving type
3. Dual-motor driving type
4. Wheel hub motor type(轮毂电机)



Lead-acid batteries(铅酸电池)

Circuit of EV(电动汽车的电路):

1. Main power (Power Battery) → motor;
2. Main power (Power Battery) → DC/DC converter → electrical equipment;
3. Braking energy recycling motor → main power (Power Battery).



12.3 Hybrid Electric Vehicle (HEV) (油电) 混合动力汽车

Vehicles that can work by two or more power sources, namely thermal power source and electric power source.

包含两种或两种以上动力源并能协调工作的车辆，即热动力源及电动力源。

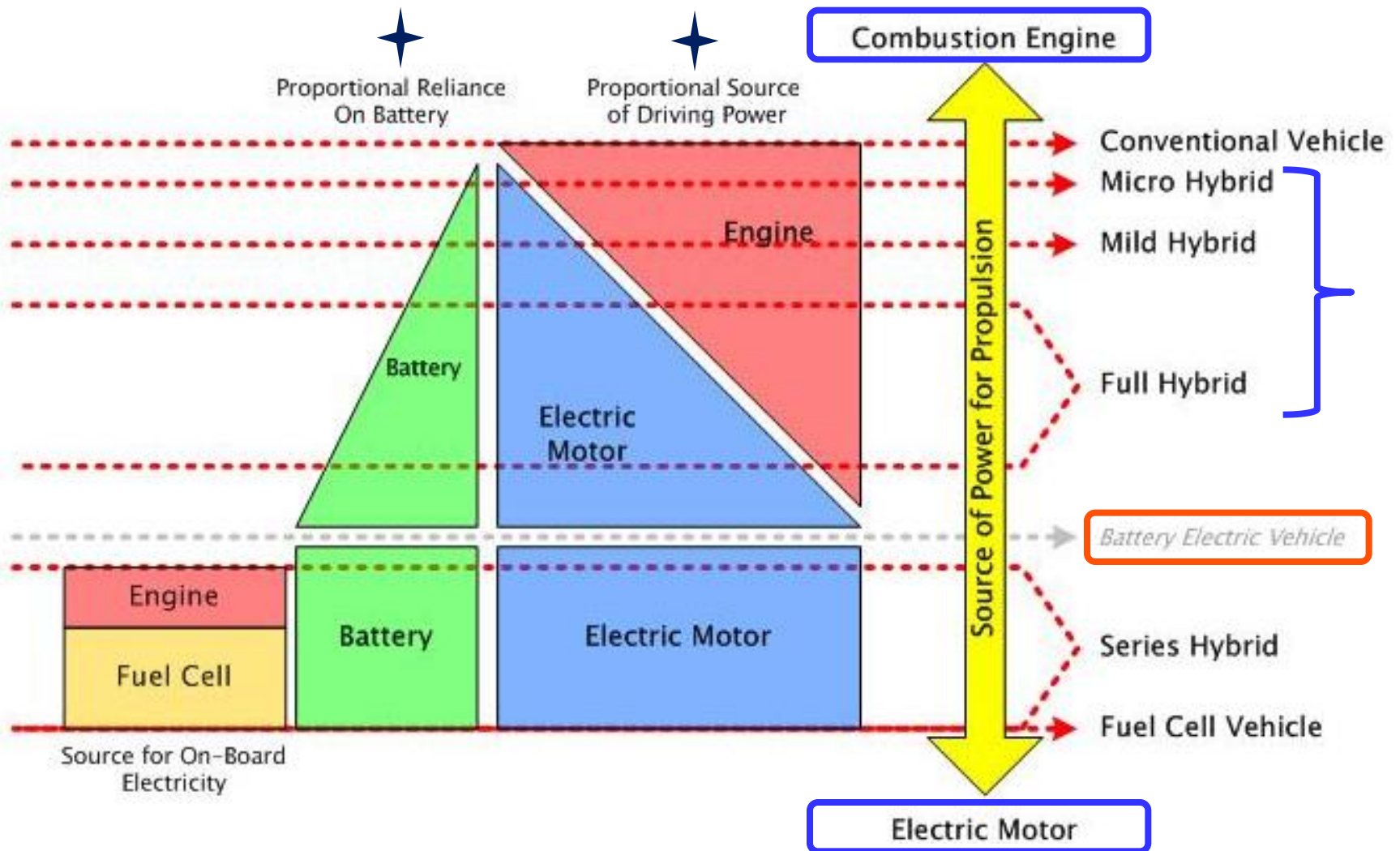


ICE

+

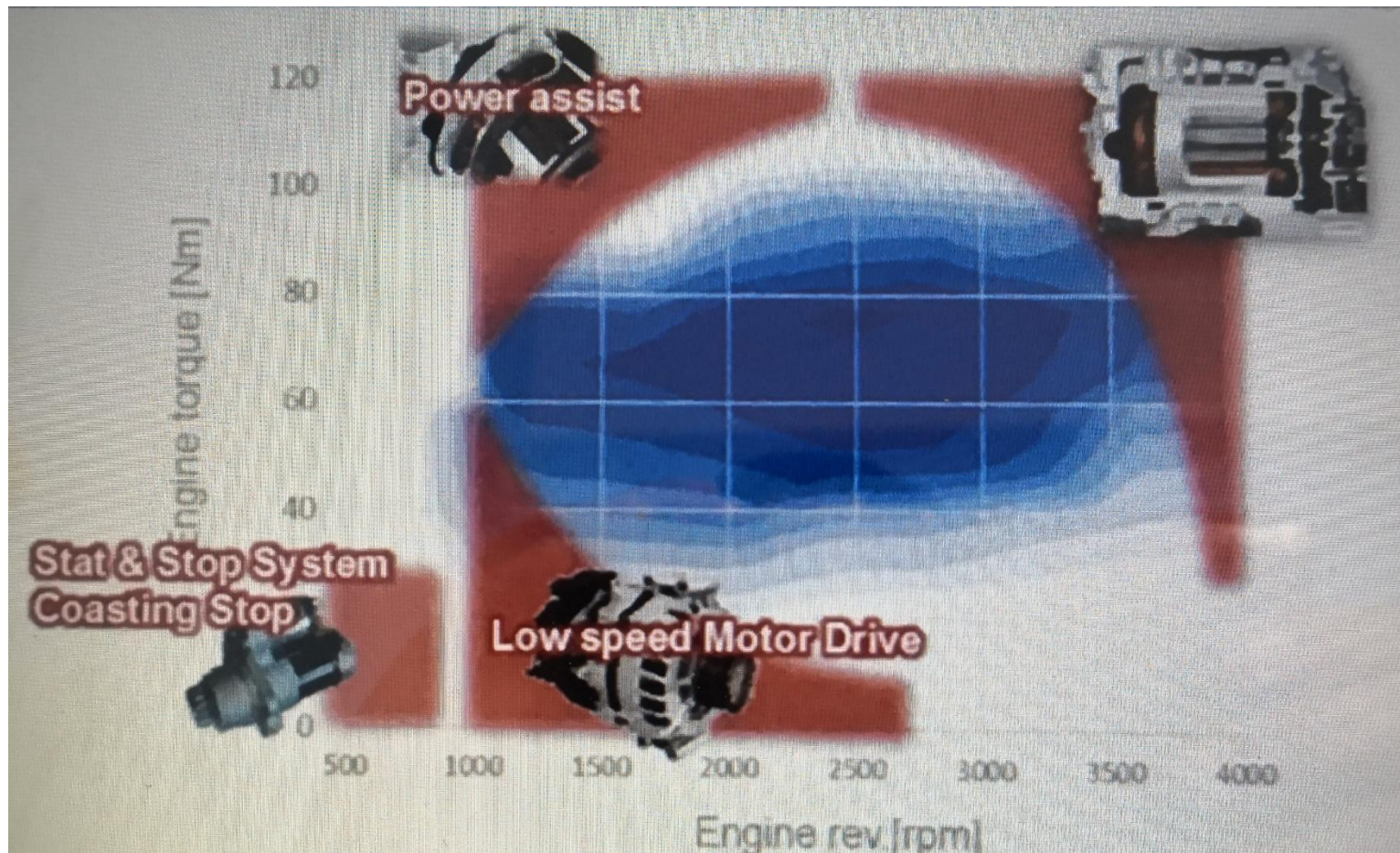
Power Battery
Ultra Capacitor

Hybrid Types



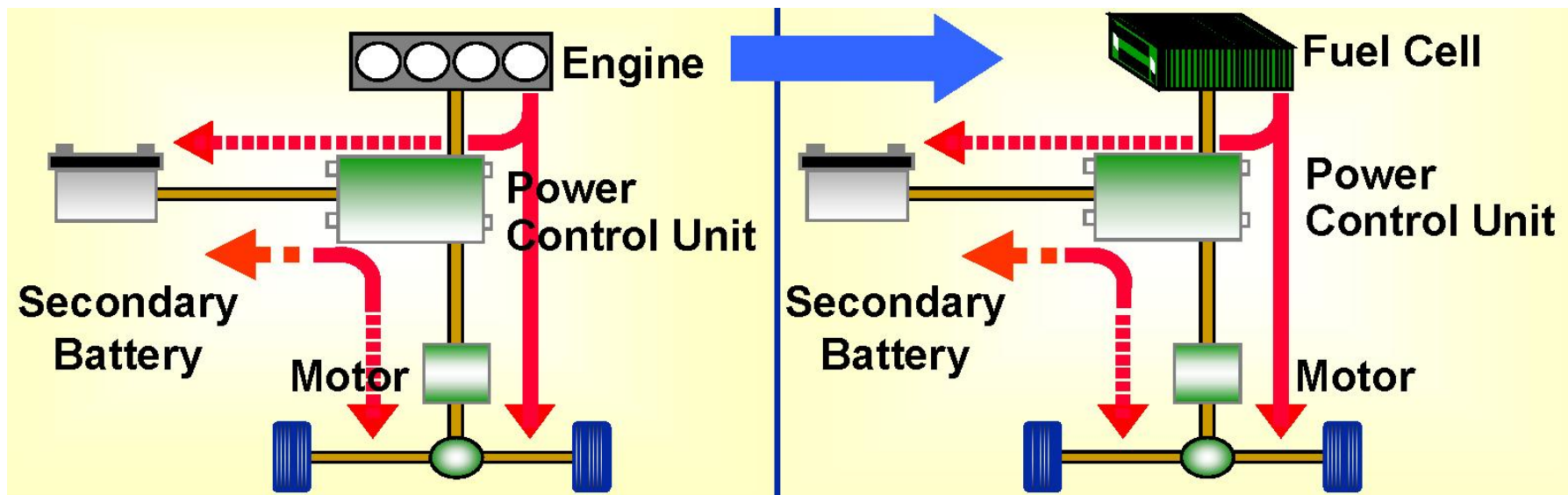
From the Green Car Congress web site: <http://www.greencarcongress.com/hybrids/>

内燃机+电机-“油电”混合动力



内燃机 + 动力电池 (HEV)

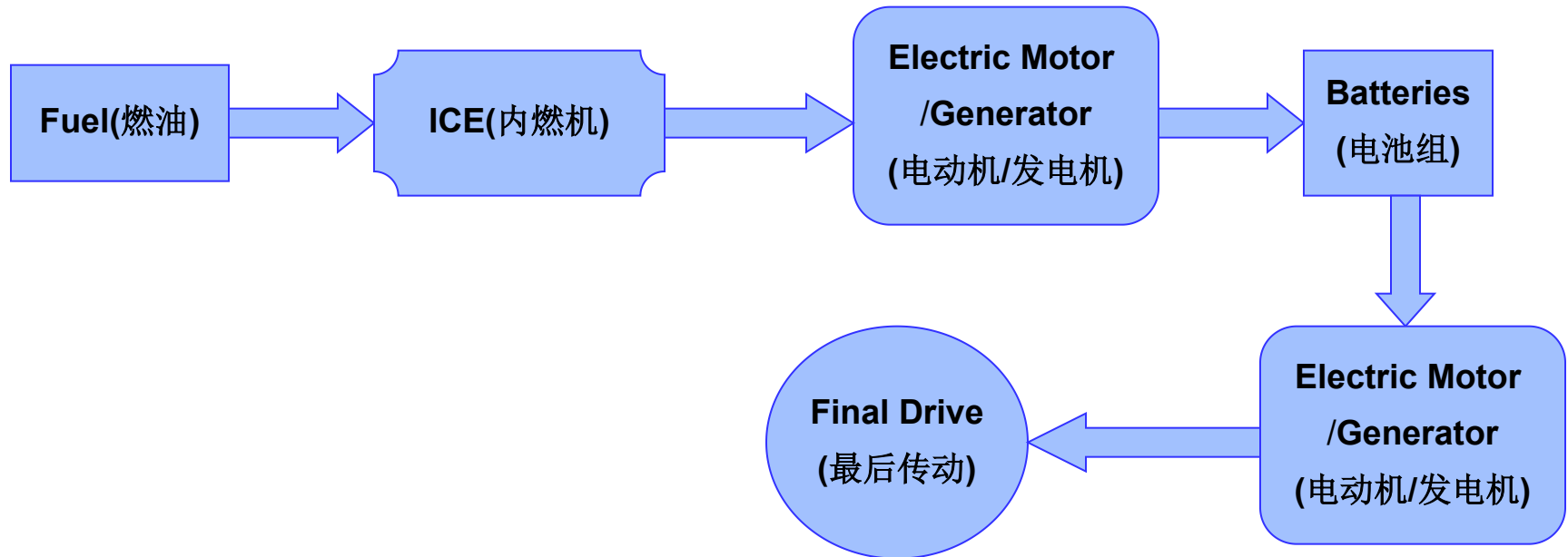
燃料电池 + 动力电池 (FCV)



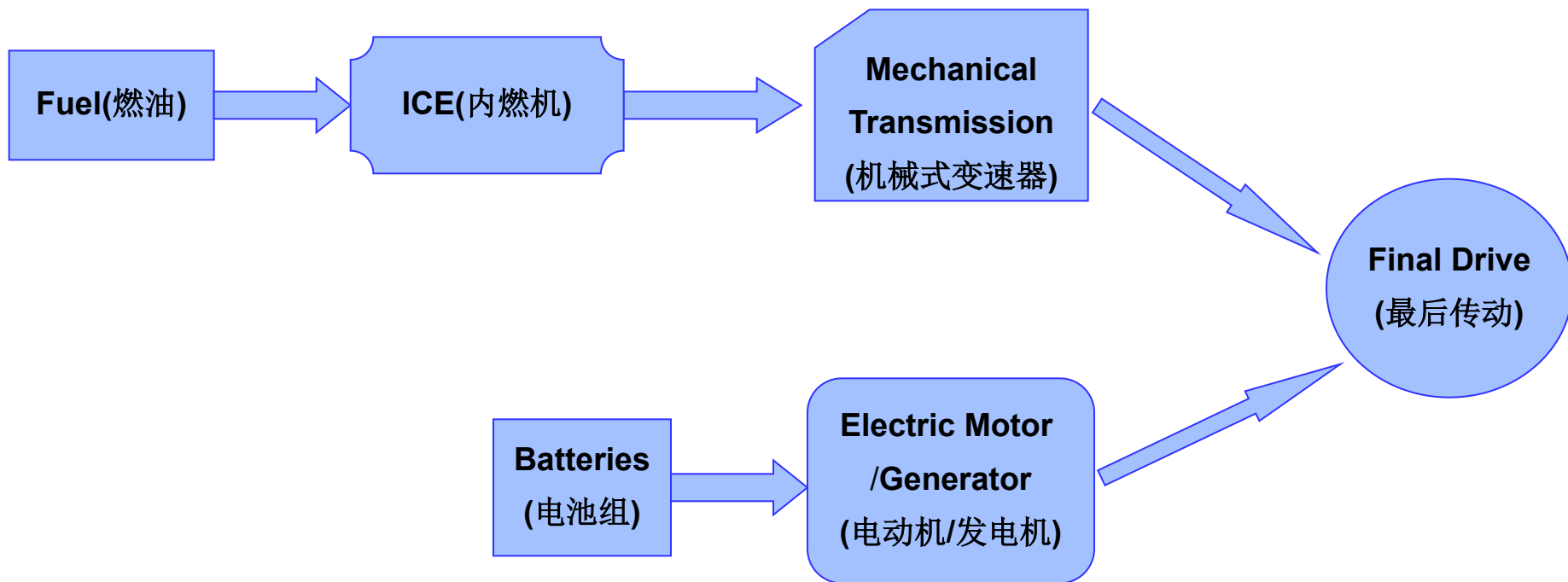
Three types based on power sources connection modes:

根据混合动力驱动的联结方式，主要分为三类：

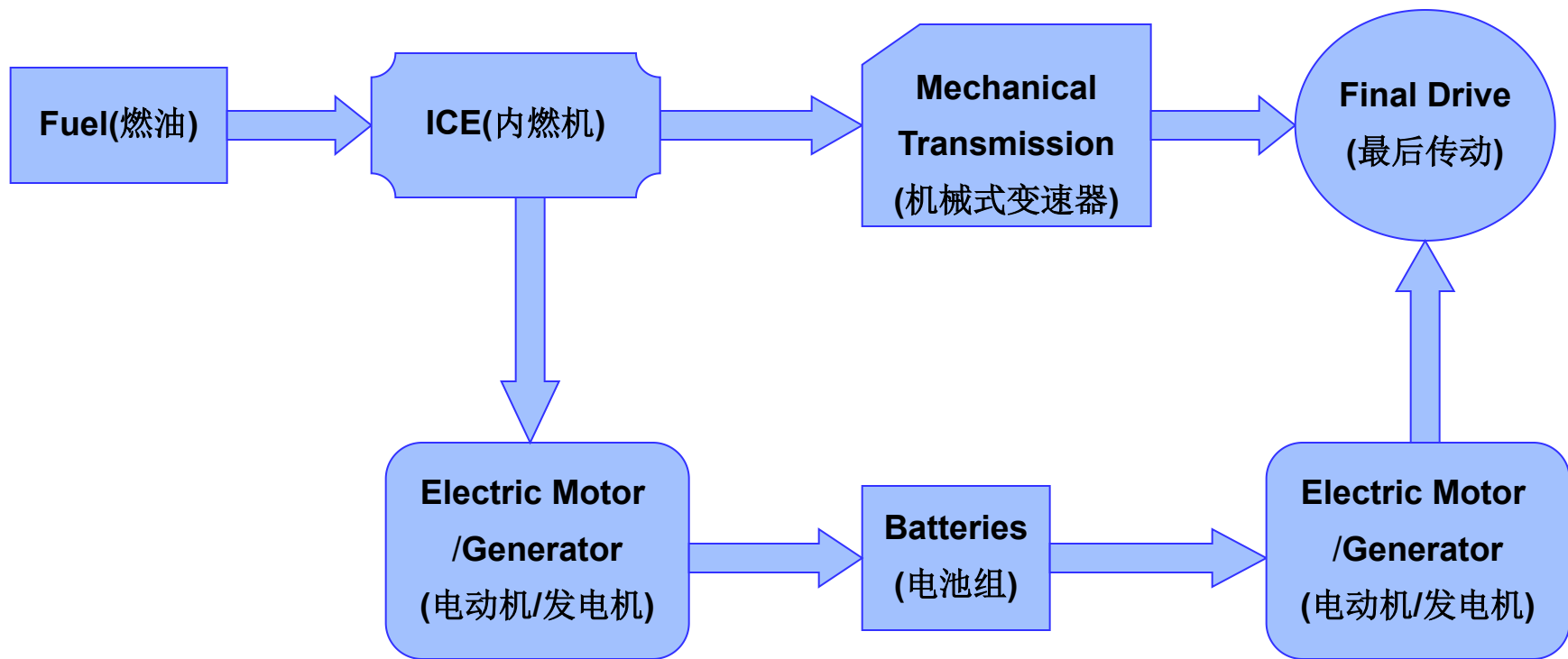
(1) Series Hybrid System (串联式混合动力系统)



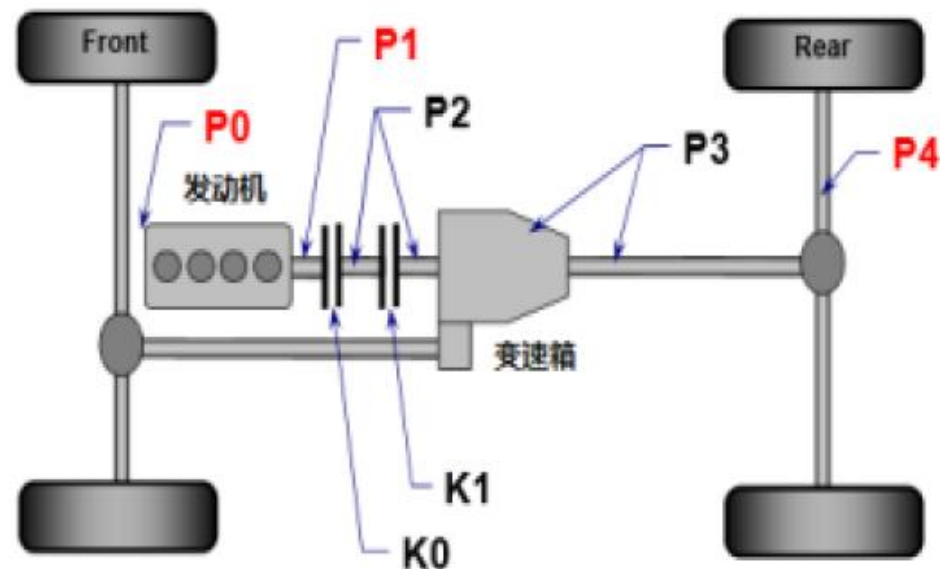
(2) Parallel Hybrid System (并联式混合动力系统)



(3) Power-split Hybrid System (混联式混合动力系统)



European styles



K0 = clutch (separation) 离合器
K1 = clutch (start-up)

- P0 = Belt driven Starter Generator (BSG)**
皮带轮
- P1 = Crankshaft mounted E- machine (CSG)**
发动机曲轴相连
- P2 = E-Machine on the gear input**
变速箱之前
- P3 = E-Maschine on the gear output**
变速箱之后
- P4 = Axle Drive**
后桥驱动

◆ THS-Toyota、iMMD-Honda、E-Power-Nissan

◆ P0、P1、P2、P3、P4

◆ Plug-in

➤ DMI-比亚迪超级混动

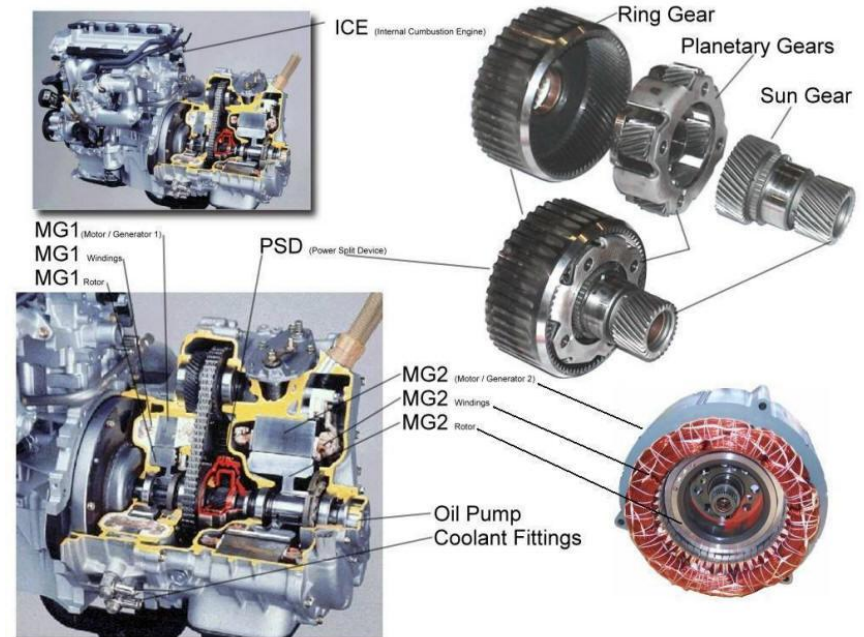
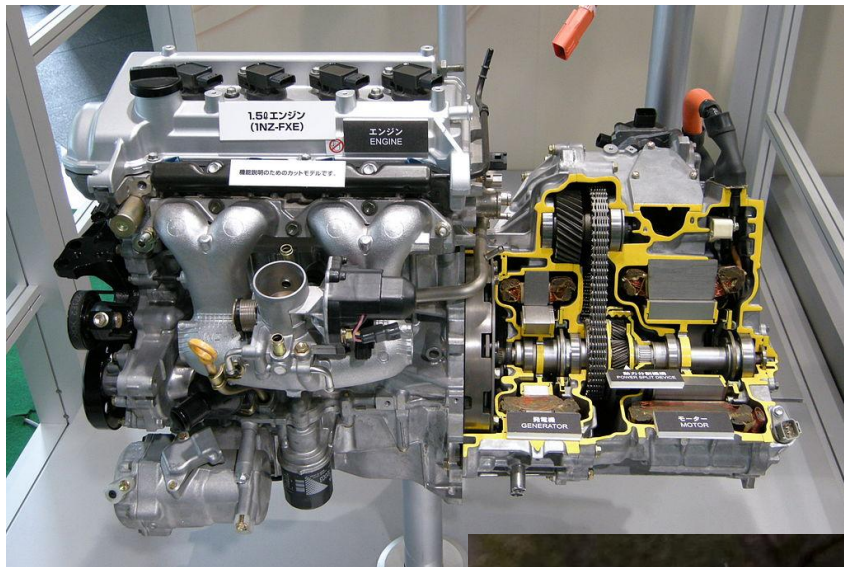
➤ HiX-吉利雷神

➤ EDU-上汽

➤ DHT-奇瑞鲲鹏

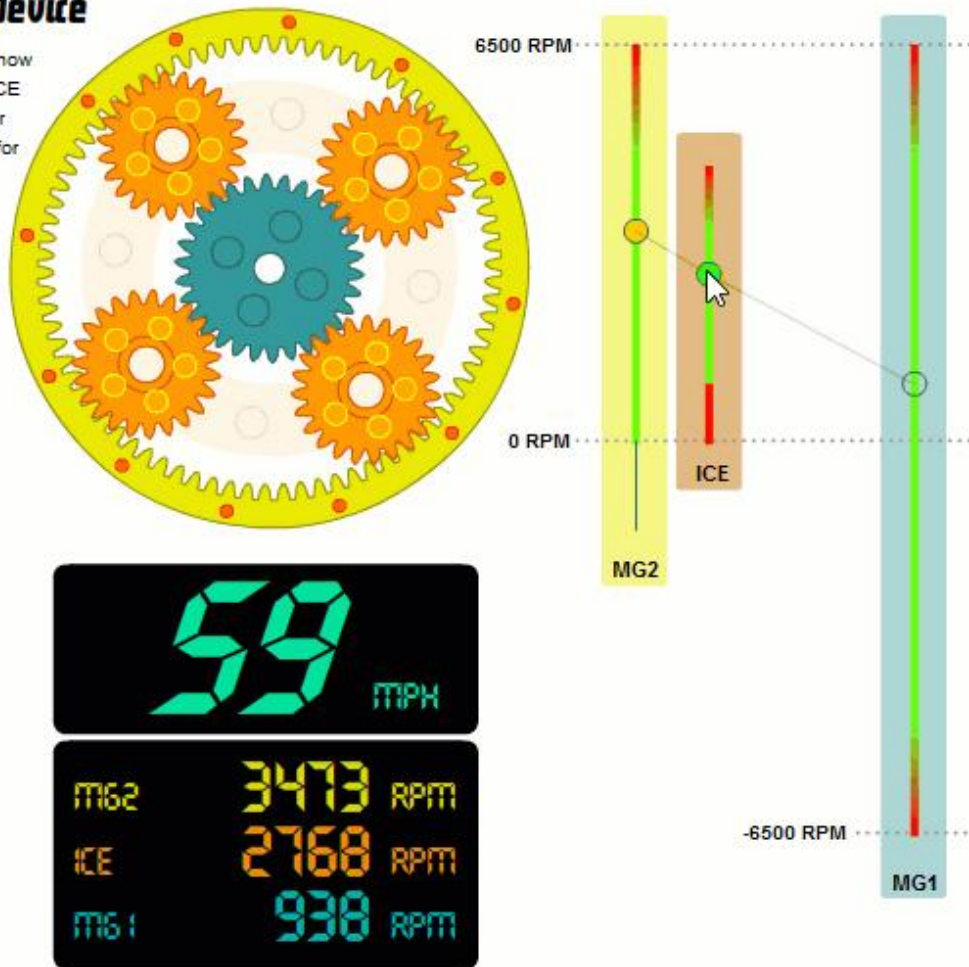
➤

丰田 THS, Toyota Hybrid System

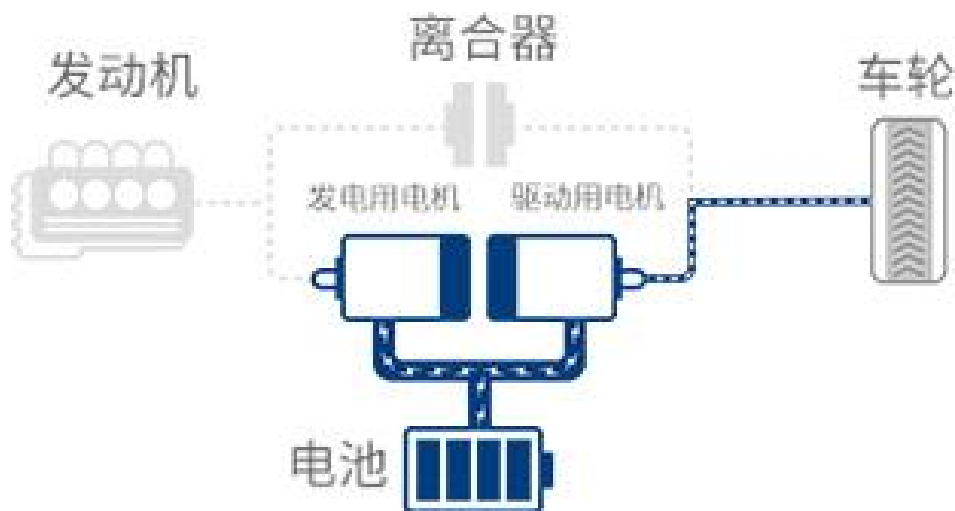


Power Split Device

Drag the sliders to see how power from MG2 and ICE is combined. Point your mouse at the diagram for



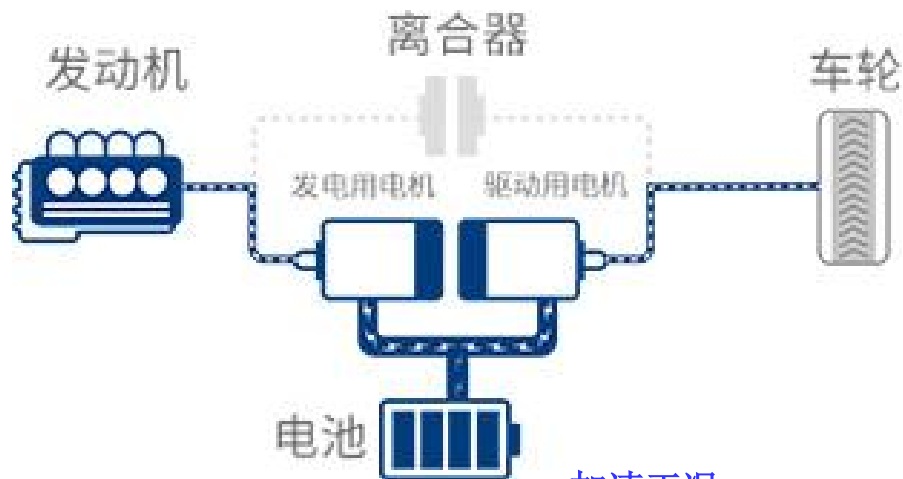
本田 i-MMD



中低速巡航工况

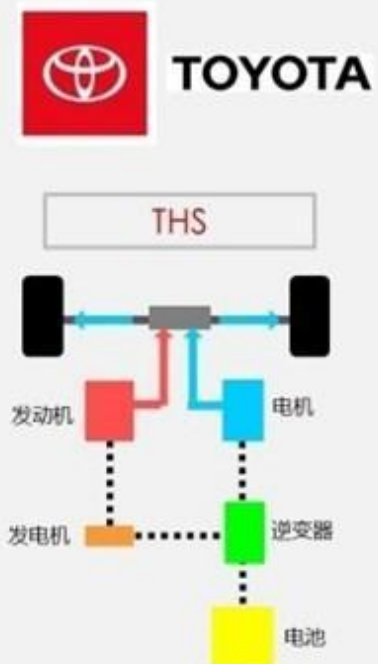


高速巡航工况



加速工况

日产 e-Power

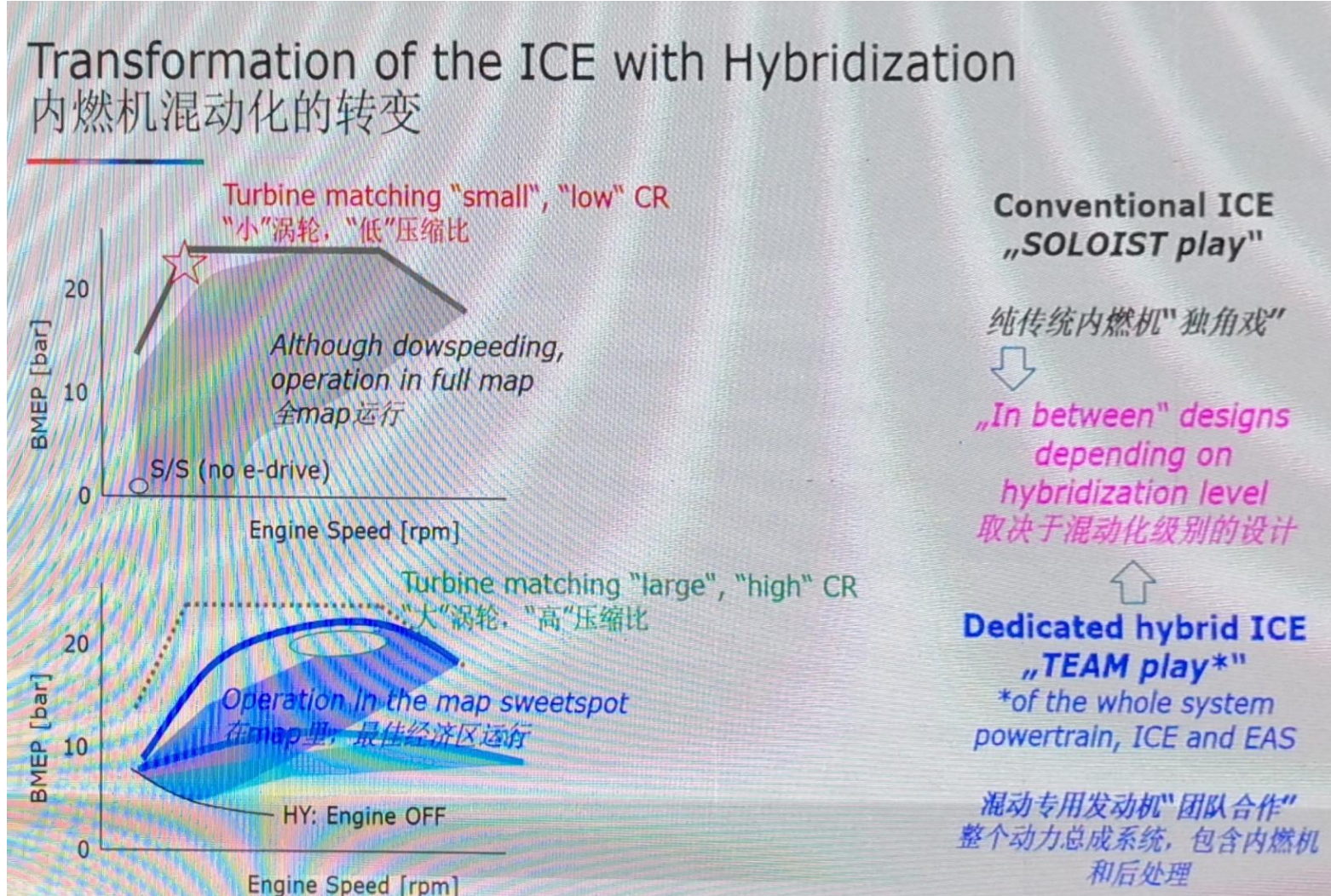


Note

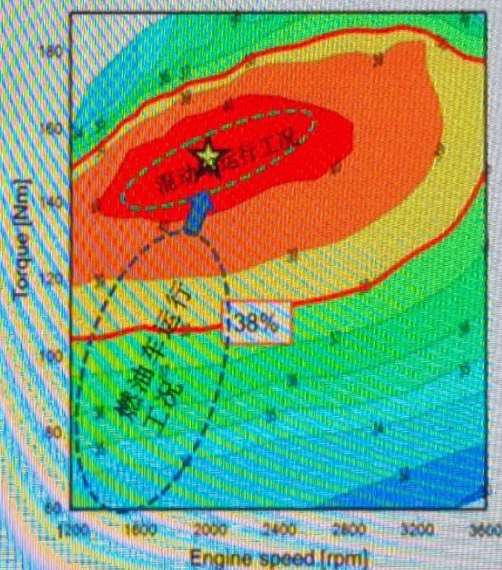


Leaf

Dedicated ICE for HEVs – DHE (Dedicated Hybrid Engine)



混合动力专用



负荷需求低:

HEV: BMEP $\sim 18\text{bar}$

常规: BMEP $23\sim 25\text{bar}$



1、减重轻量化:

- 曲轴轴颈减小;
- 连杆活塞减轻优化;
- 机体缸盖减重优化;
- 取消前端附件;
- 同排量4缸 $\rightarrow$ 3缸;

2、降低成本:

- 可变VVL和VVT $\rightarrow$ 固定气门相位;
- 滚子摇臂 $\rightarrow$ 杯状挺柱驱动;
- 可变排量机油泵 $\rightarrow$ 固定机油泵;
- 取消可变滚流阀
- 取消可变进气歧管长度

◆ ICE special for HEVs

12.4 Fuel Cell Vehicles

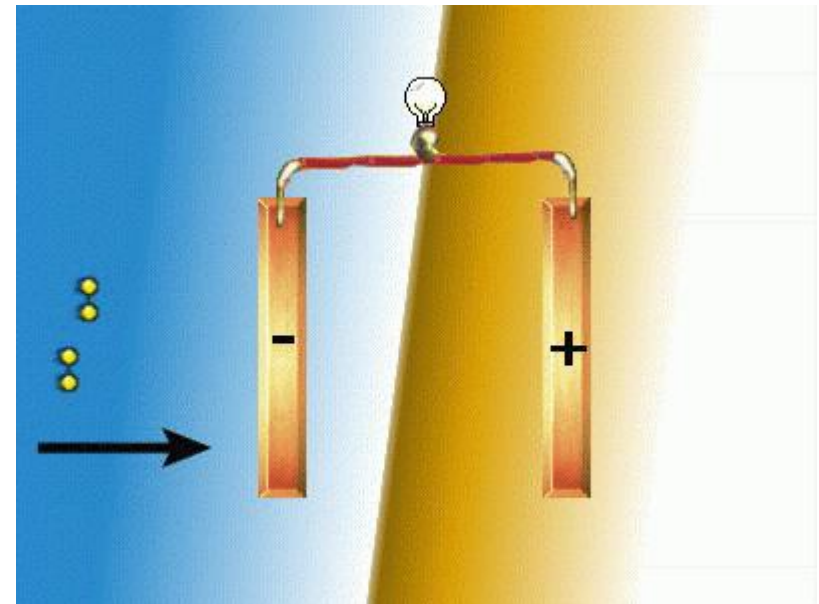
(燃料电池汽车)

- **Fuel Cell**

A device that converts the chemical energy from a fuel to electricity by a catalytic chemical reaction with oxygen.

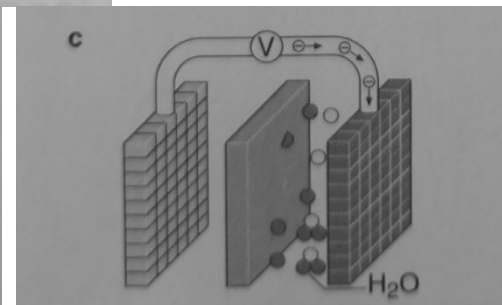
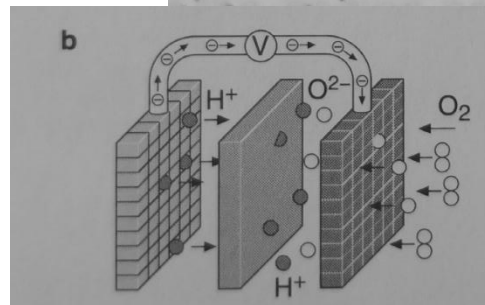
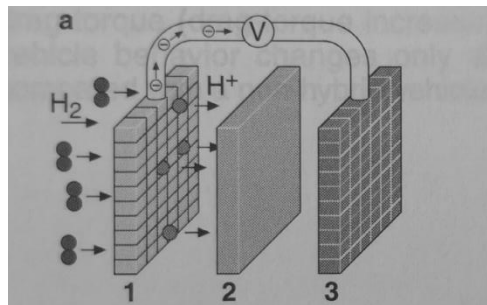
- **Working principle**

- ✓ Anode (阳极) $2H_2 \rightarrow 4H^+ + 4e^-$
- ✓ Cathode (阴极) $4H^+ + 4e^- + O_2 \rightarrow 2H_2O$
- ✓ Overall $2H_2 + O_2 \rightarrow 2H_2O$
- ✓ Produce direct current (直流电)
- ✓ Classified by the type of electrolyte (电解质) used



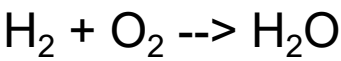
a) Hydrogen oxidation,
b) Oxygen reduction,
c) Water production.

1 Anode,
2 Membrane,
3 Cathode.

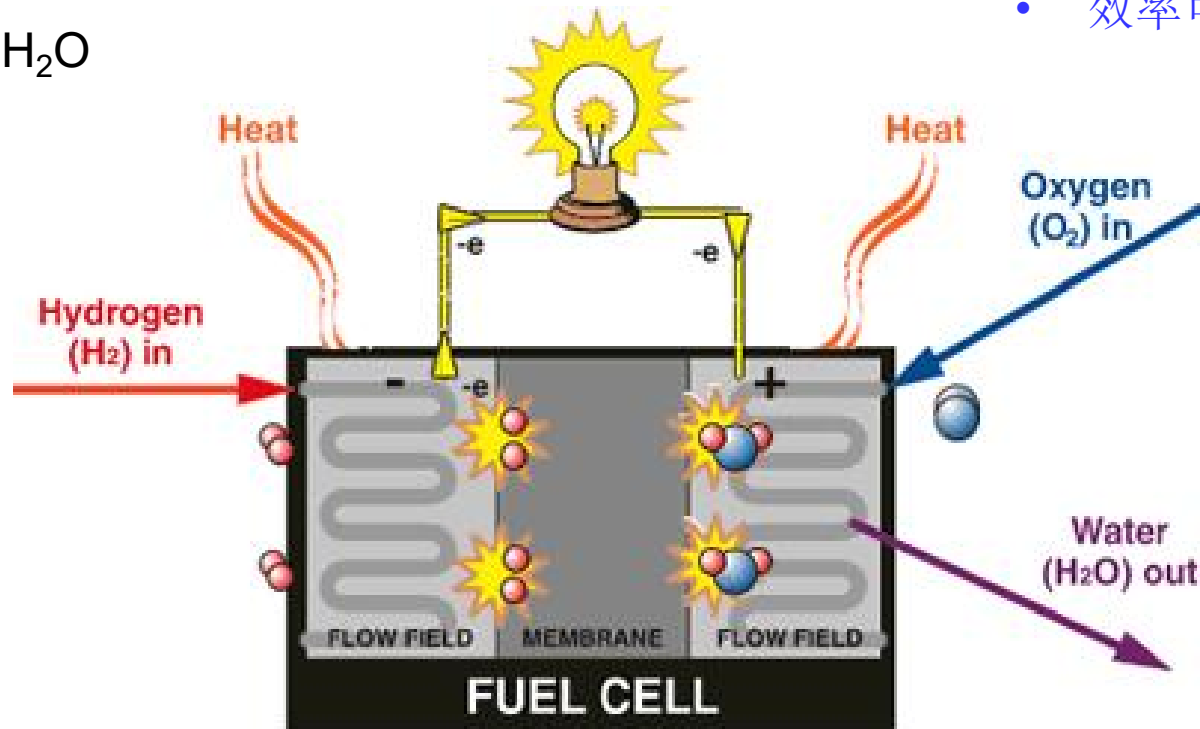


车用质子交换膜燃料电池

Overall Reactions:

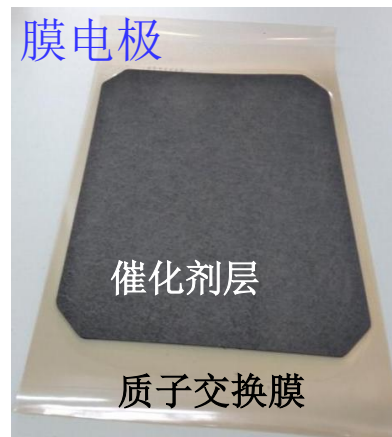
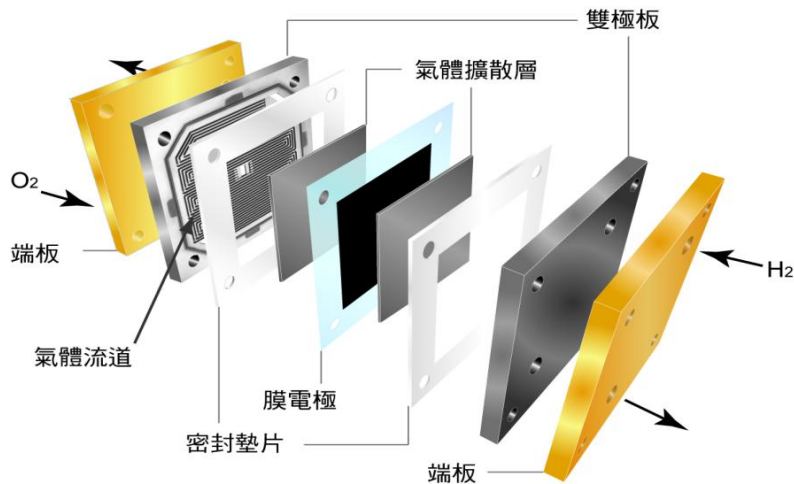
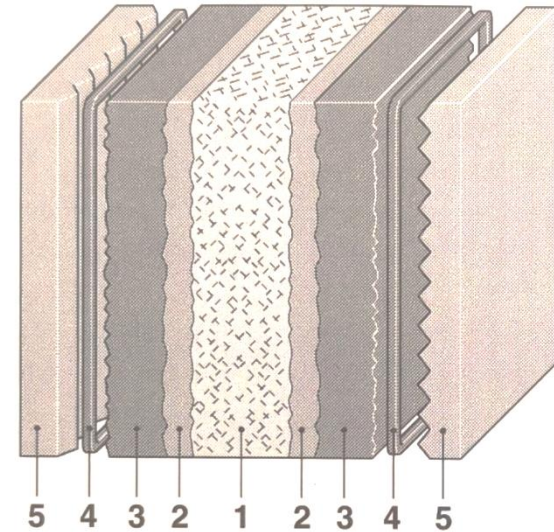
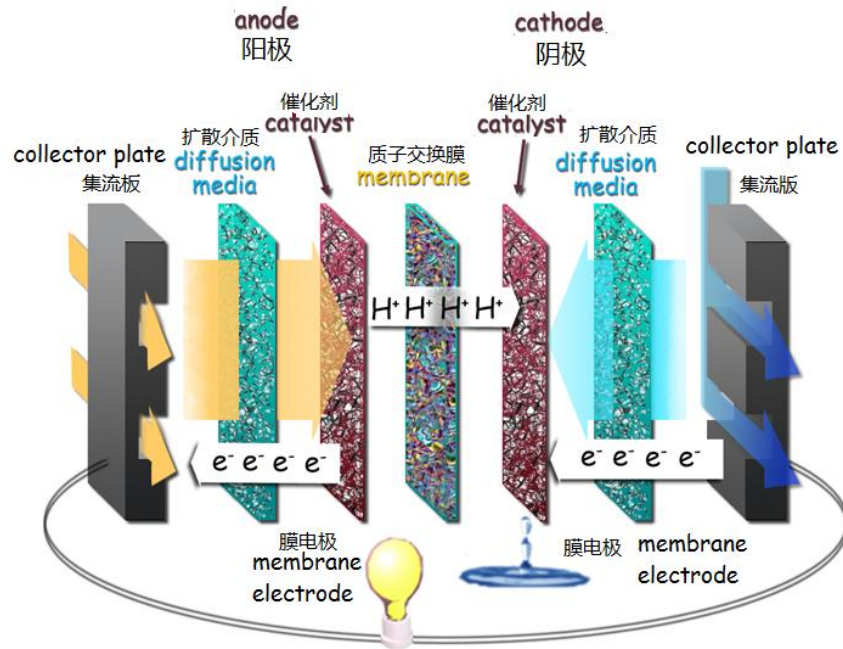


- 非燃烧的电化学反应
- 产物为电、水和热量
- 效率可>50%



Structure of Fuel Cell

- 1 Membrane (electrolyte layer),
- 2 Electrodes (catalyst layer),
- 3 Gas-diffusion layers,
- 4 Seals,
- 5 Bipolar plates.



单电池特性

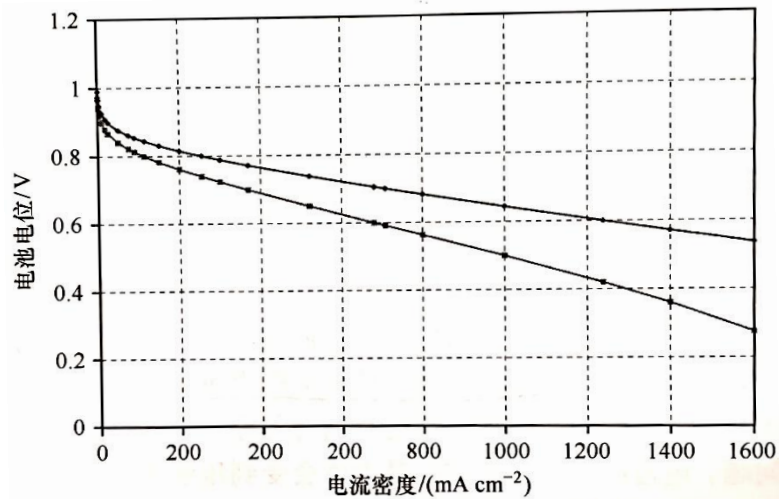
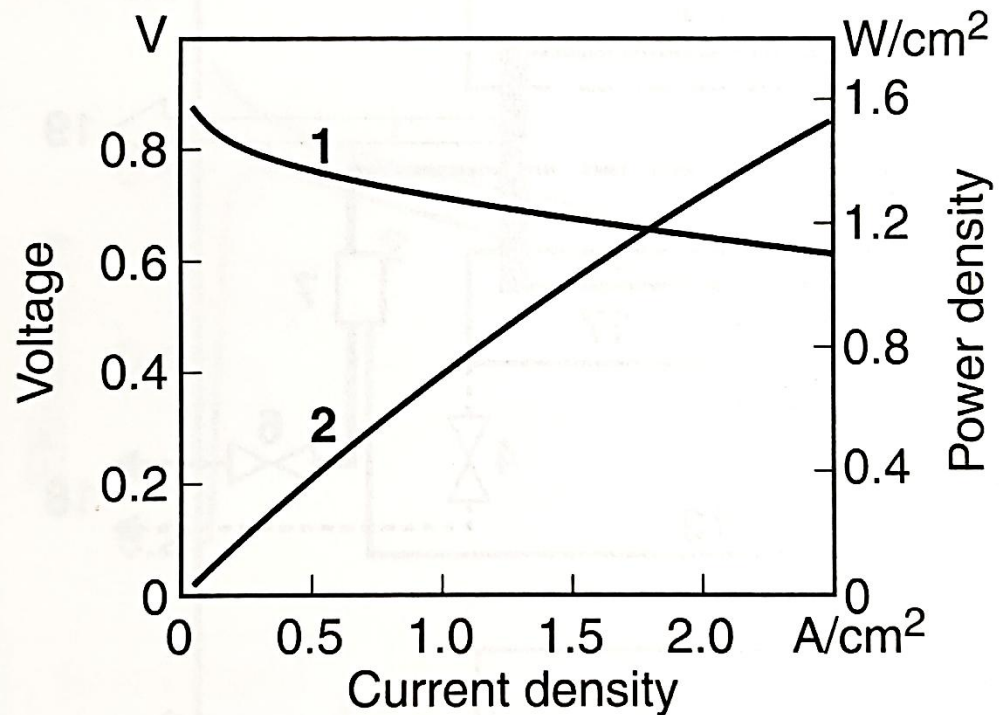


Figure 3: Electrical characteristic quantities of the fuel cell

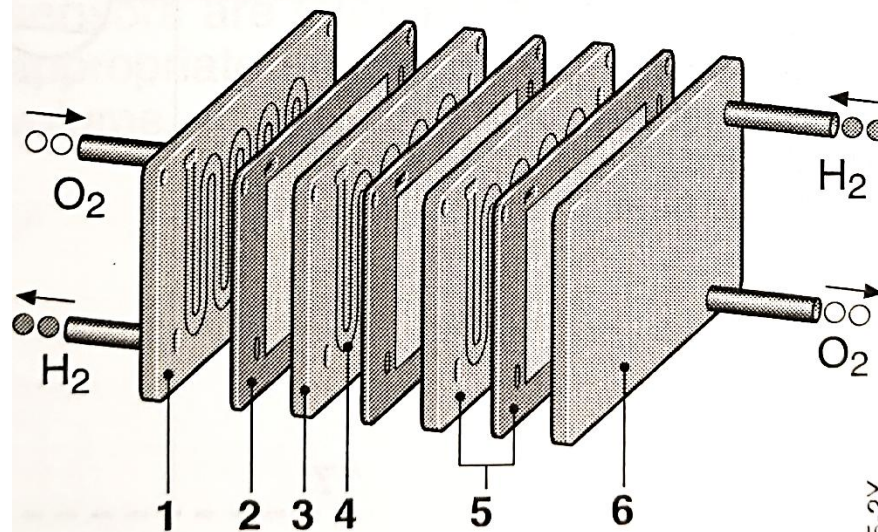
1 Voltage,
2 Power density.



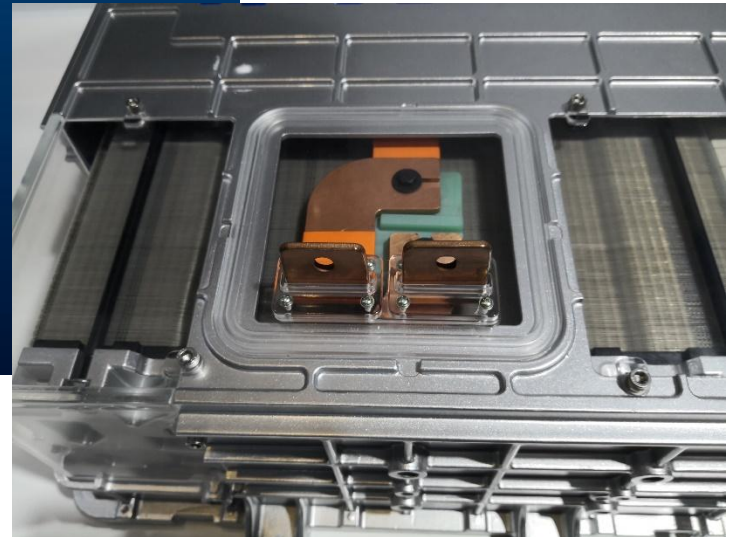
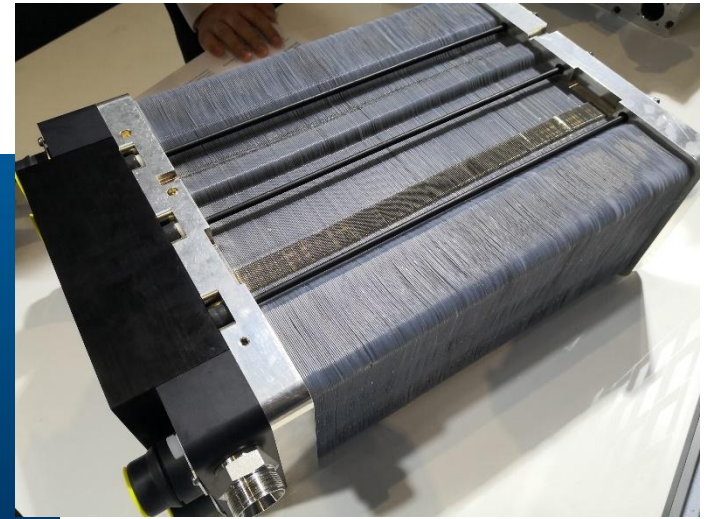
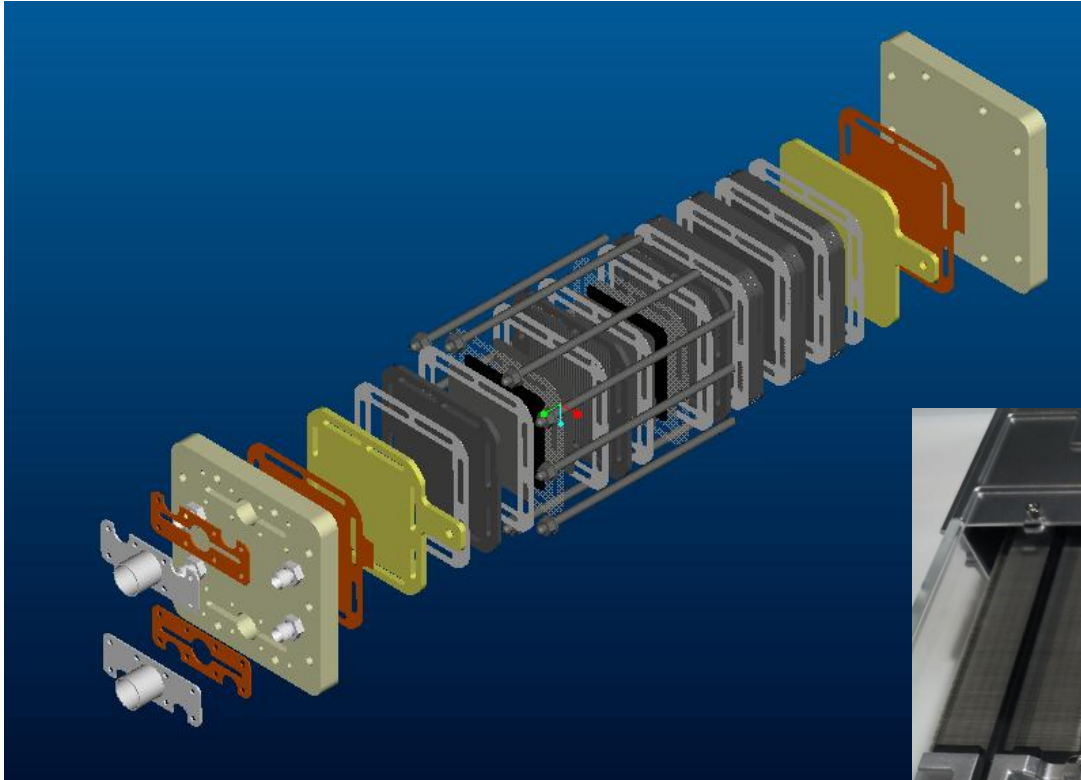
单电池→电池堆

Figure 4: Structure of a fuel-cell stack

- 1 End plate
- 2 Membrane electrode unit with membrane, catalyst material and seals
- 3 Bipolar plates
- 4 Gas channels
- 5 Repeating units
- 6 End plate



STB0015-2Y



燃料电池发动机

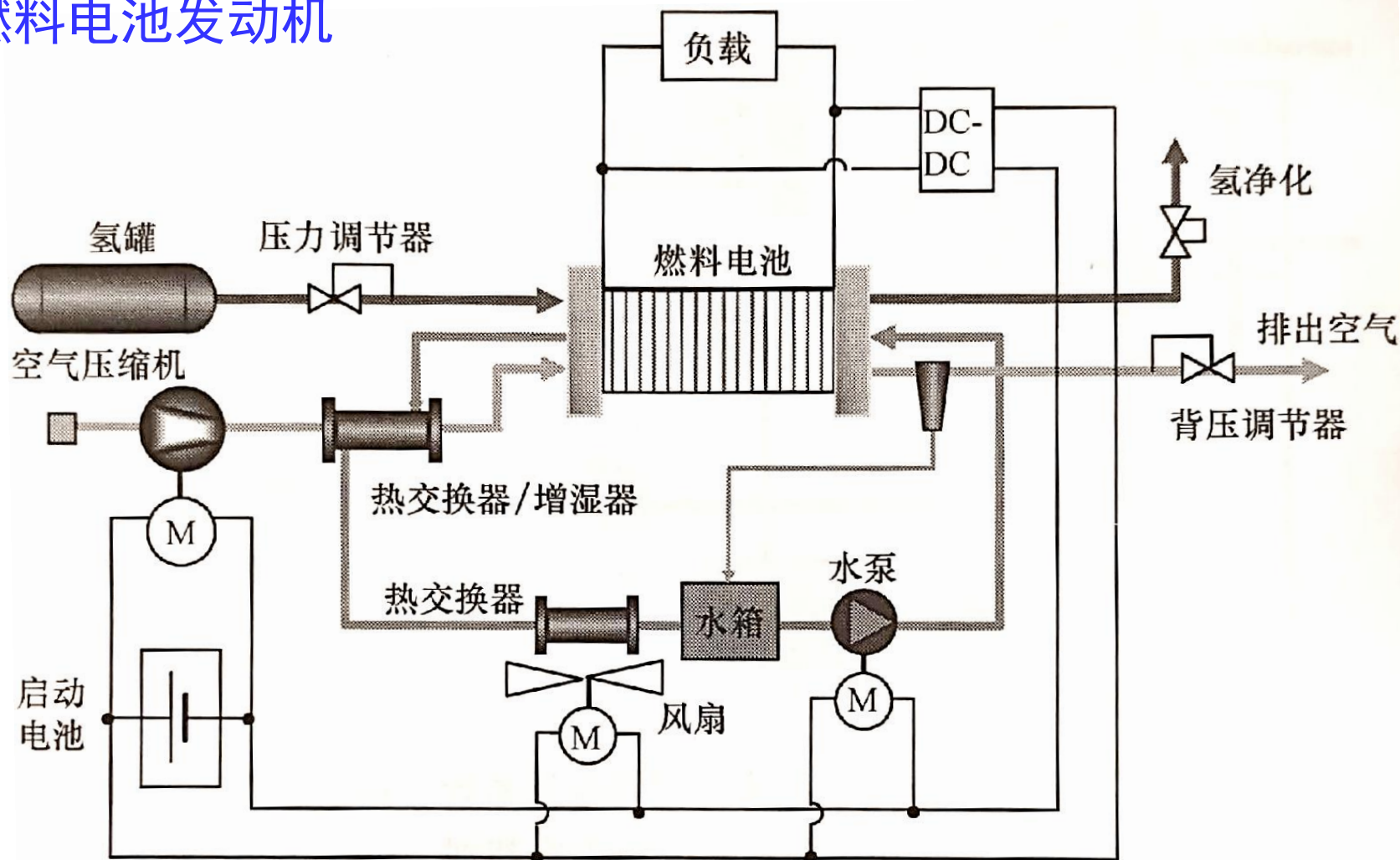
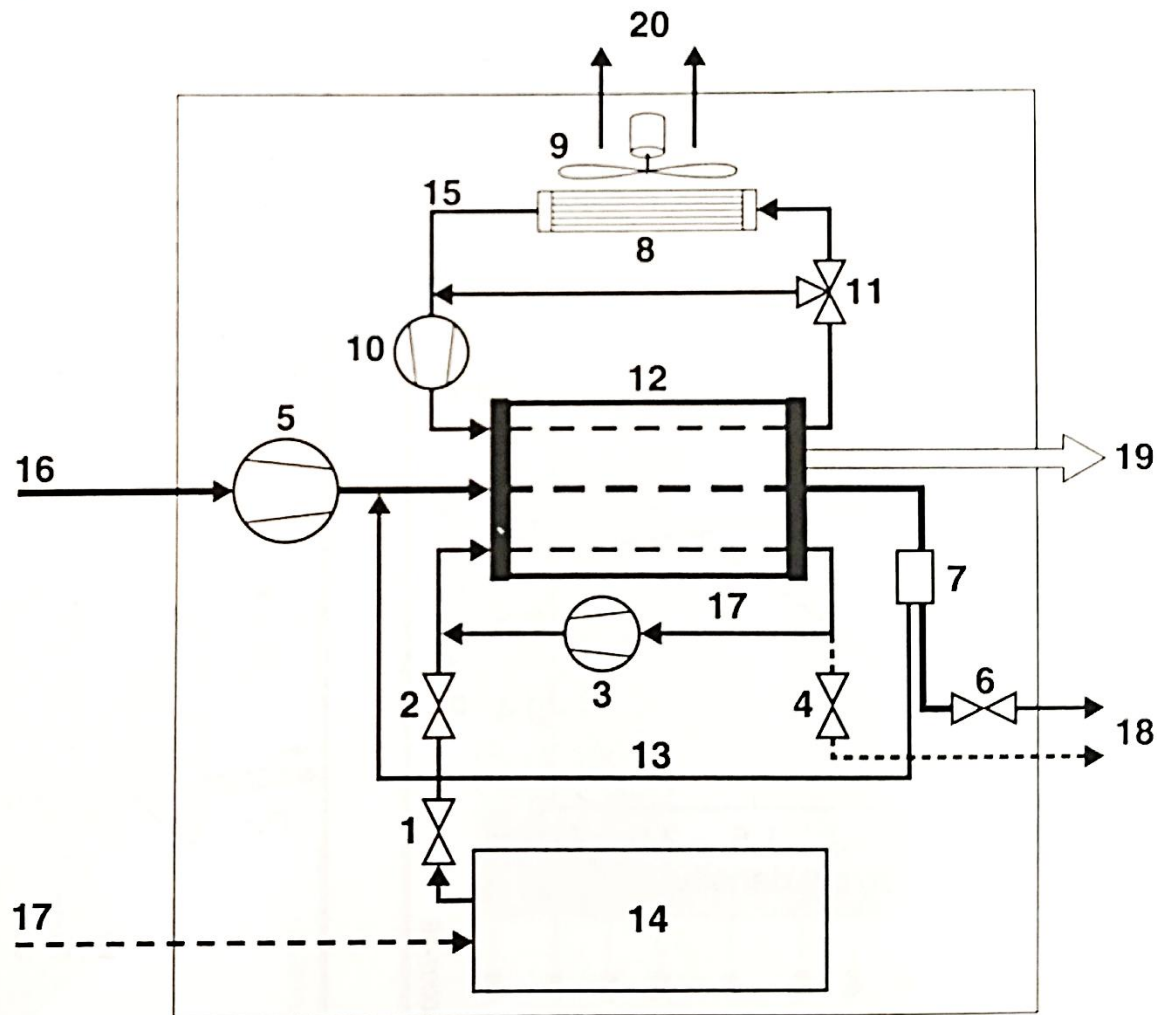
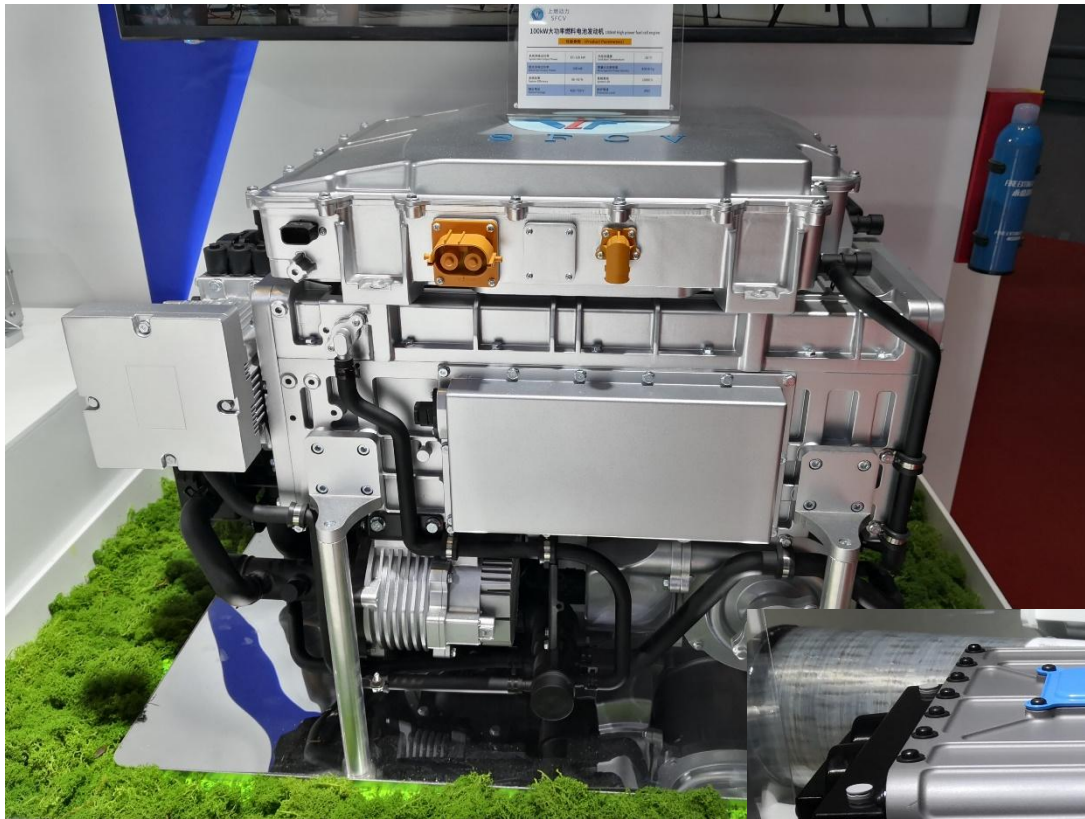


Figure 5: Fuel-cell system

- 1 Pressure reducer for hydrogen,
- 2 Hydrogen metering valve,
- 3 Hydrogen recirculation fan,
- 4 Drain valve,
- 5 Air compressor,
- 6 Dynamic-pressure control valve,
- 7 Water management,
- 8 Radiator,
- 9 Radiator fan,
- 10 Coolant pump,
- 11 Coolant control valve,
- 12 Fuel cell stack
- 13 Water
- 14 Hydrogen tank
- 15 Coolant
- 16 Air
- 17 Hydrogen
- 18 Exhaust gas
- 19 Electric power
- 20 Waste heat



STB0013-3E



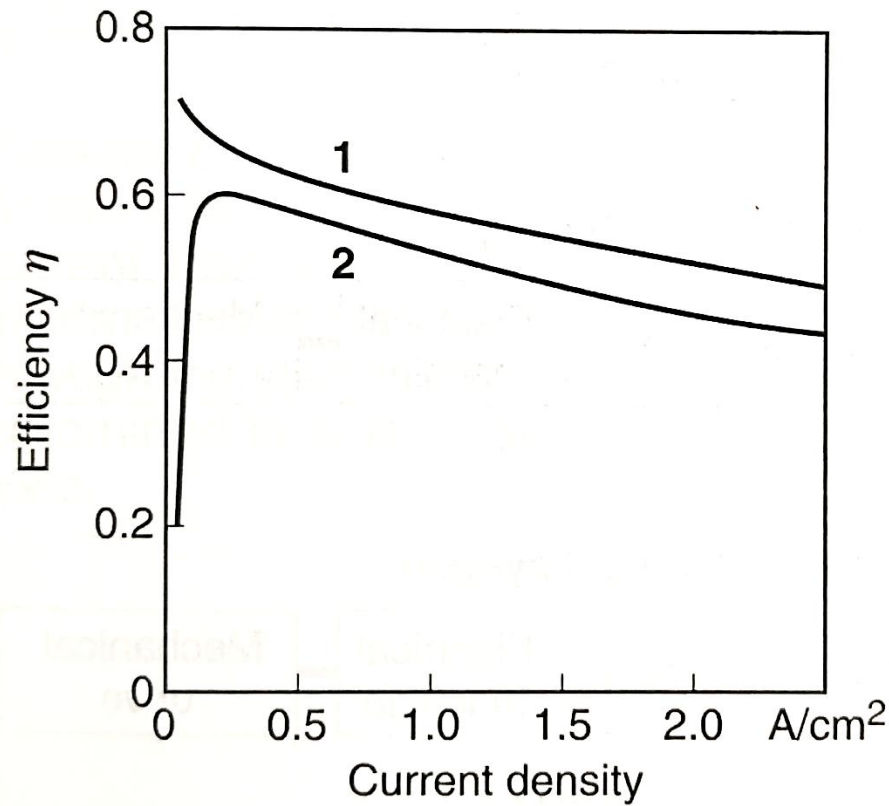
净功率:
108kW (10-110kW)
系统效率: 56-42%



净功率:
75kW

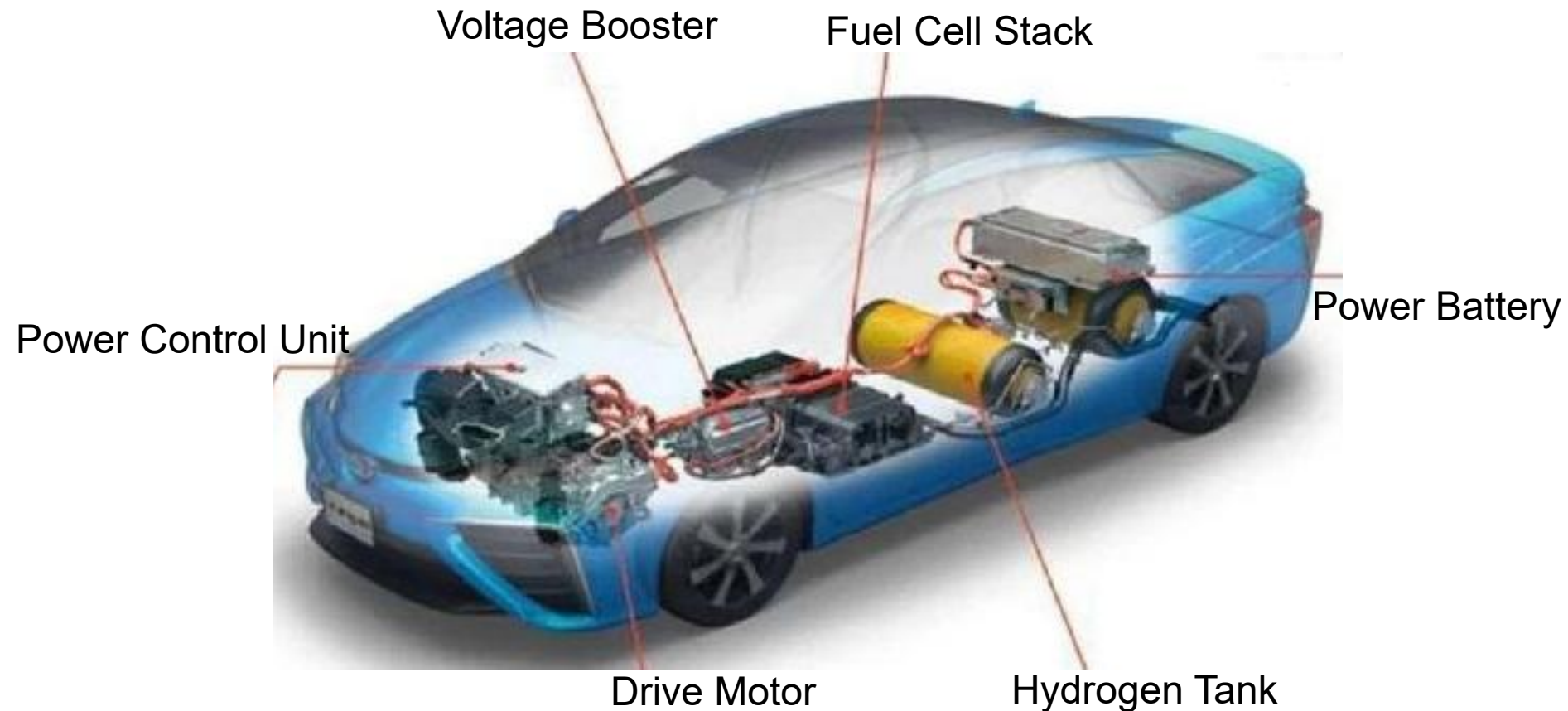
Figure 6: Efficiency of fuel-cell stack and fuel-cell system

1 Efficiency of fuel-cell stack,
2 Efficiency of fuel-cell system.

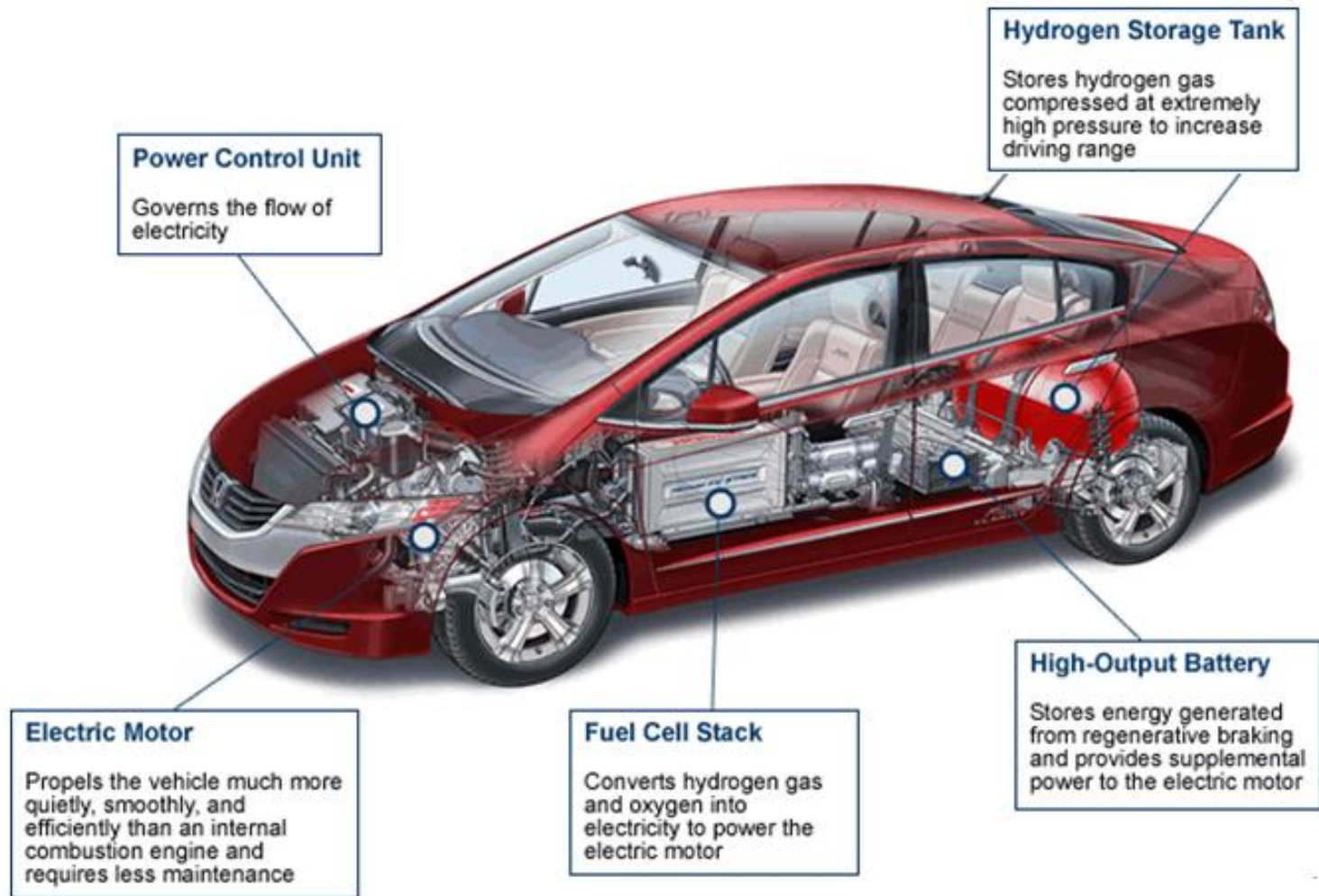


STB00008-4E

燃料电池汽车（Mirai）



“燃料电池”必须组成“动力系统”才能适用于汽车
“燃料电池”+“动力蓄电池”+“电机”



- FCEV in Tongji University



Challenges for hydrogen FCEVs



- ✓ High production cost, 99.999% H₂
- ✓ Lack of extensive hydrogen infrastructure
- ✓ H₂ safety



- ✓ Lower energy density in volume
- ✓ Large amount of energy required to isolate hydrogen from natural compounds, like H₂O

Transportation needs special kind of energy source

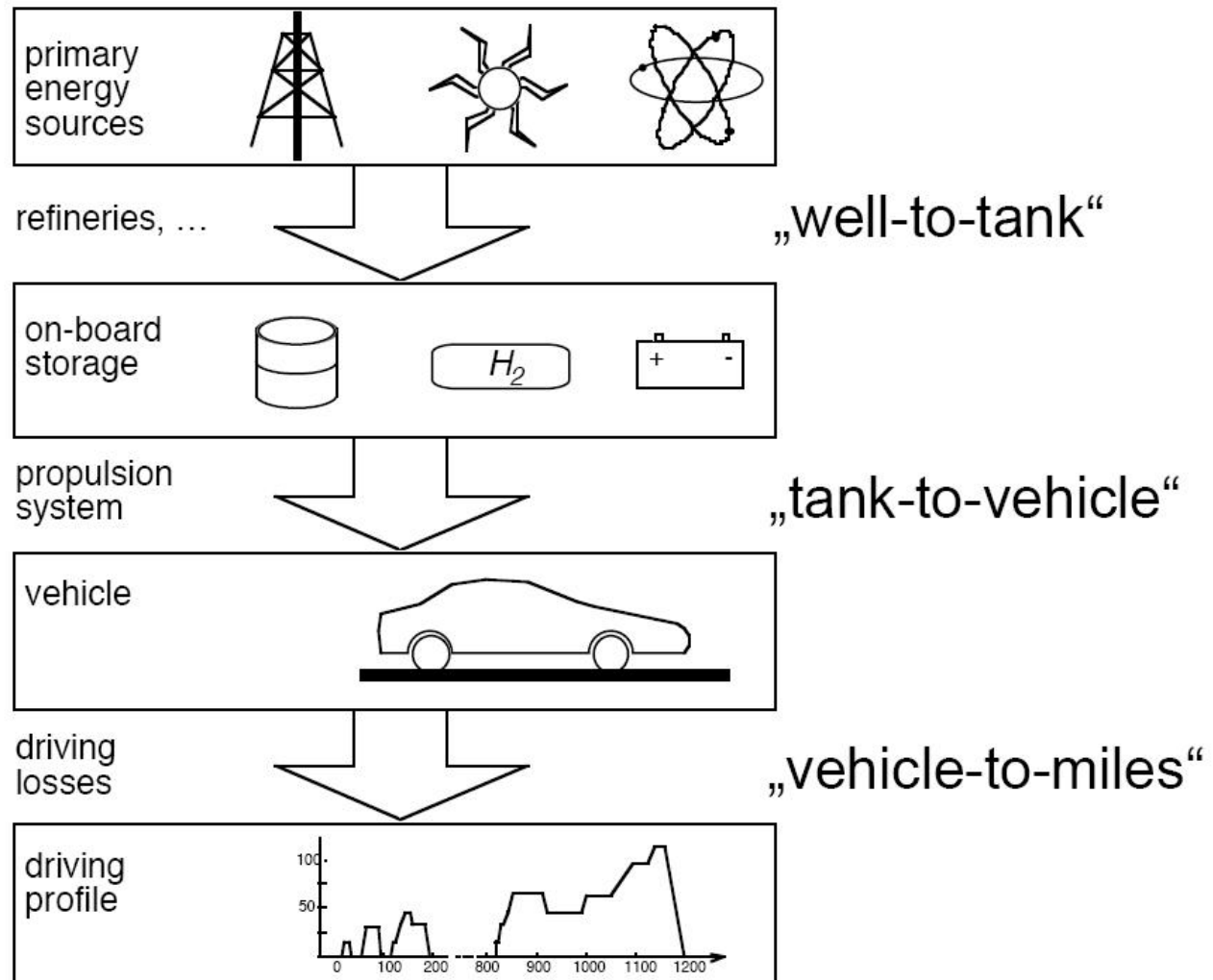
- Vehicles need to carry source of energy on board
- Hydrocarbons are unparalleled in terms of energy density
 - For example, look at refueling of gasoline
 - ~40 Liters in 2 minutes (~0.25 kg/sec)
 - Corresponding energy flow
 - = 0.25 kg/s * 44 MJ/kg
 - = 11 Mega Watts

Liquid hydrocarbons !

12.5 双碳背景下的新型交通“燃料”

源于太阳能、风能等的二次能源

Well to wheel – whole life cycle



Source : Guzzella, ETH Zurich (2006)

“miles-to-CO₂”

“绿电”  : 太阳能/风能
 电网

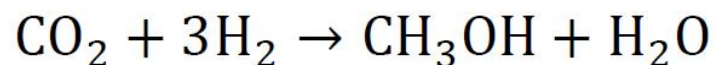
“绿氢” H_2 : 最小分子气体; 输运难; 易爆炸

太阳能/风能—电 $\rightarrow\rightarrow$ 电解水— H_2

“绿醇”——甲醇 CH_3OH : H质量占比12.5%; 常压液态; 低毒性

太阳能/风能—电—电解水— H_2

碳 (C) 来自空气、碳捕集、高 CO_2 排放工业



催化剂: CuO 、 ZnO 、 MgO 、 Al_2O_3 ; 工作温度: $>200^\circ C$ 。

“绿氨” NH_3 : H质量占比17.6%; 高压/低温液态; 低毒性

太阳能/风能—电—电解水— H_2

氮 (N) 来自空气中的 N_2



能源供给侧

储存

运输

能源消费侧

一次能源

二次能源

蓄热

蓄水

蓄气

盐穴
压缩
空气

蓄电

电池

原料
燃料

液体
气体

电网

管道
罐

电机

电炉

电池

化工

炼炉

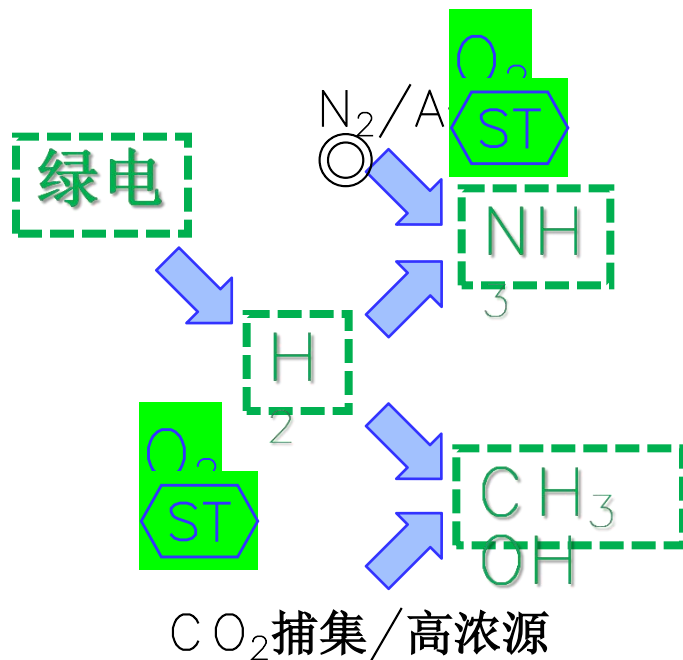
燃气轮机

内燃机

燃料电池

风能
太阳能
生物质能

分布式
ST



时空边界：全生命周期

竞争点：无碳、高效、无害、低社会成本

电/氢/甲醇/氨的利用

1、电

→高容量/高功率动力蓄电池

→高功率密度电机

“新能源”汽车

2、内燃机（高温利用）

→直接燃烧，掺烧

3、燃料电池（低温利用）

→直接利用（氢/甲醇/氨燃料电池）

→间接利用（甲醇/氨重整制氢+氢燃料电池）

性能	单位	氢	氨	甲醇	乙醇	汽油	柴油	甲烷
密度	g/cm ³	0.071	0.77（液）	0.281	0.789	0.73	0.84	0.187
粘度(25℃)	mPa·s	8.8e-3	1.007e-2	0.545	1.074	0.6	1.352	1.107e-2
燃点	℃	400	800	473	363	530	220	650
汽化潜热	kJ/kg	//	1370	1165	953	275	232	//
饱和蒸汽压 (25℃)	kPa	//	1003.2	13.0	6.0	//	//	//
低热值	MJ/kg	120	18.6	19.9	26.8	43.5	42.7	38.1
辛烷值	RON	130	110	119	108.7	90~98	25-30	107
爆炸极限	%vol.	4.5~75	16~28	6.0~36.5	3.3-19	1.4~7.6	0.7~5.0	5~15
能量密度	MJ/m ³	3.75	11.3	15.6	21.1	31	35	7
最小点火能量	mJ	0.02	680	0.14	0.24	0.2	0.63	0.32
火焰传播速度	m/s	3.51	0.07	0.36	0.43	0.58	0.13	0.38
沸点	K	20	239	337.8	351	310-477	453-683	112
凝点	K	14	195	175.6	159	198	229-281	90
比热比	γ	1.41	1.31	1.20	1.13	1.15	1.15	1.32